Aga Khan University Examination Board

Notes from E-Marking Centre on HSSC-II Mathematics Examination May 2017

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC-II) Mathematics. It contains comments on candidates' responses to the 2017 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses which support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that require candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

Key observations:

This year candidates did not perform well in questions based on linear programing, applications of differentiation and integration, transformation, and coordinate geometry. Candidates had problem in converting tabular data into meaningful mathematical equations according to the given situation and working with tangents to circle. Candidates performed well on questions related to complex numbers, functions, partial fractions, parabola, ellipse, and hyperbola.

Detailed Comments:

Question 1i:

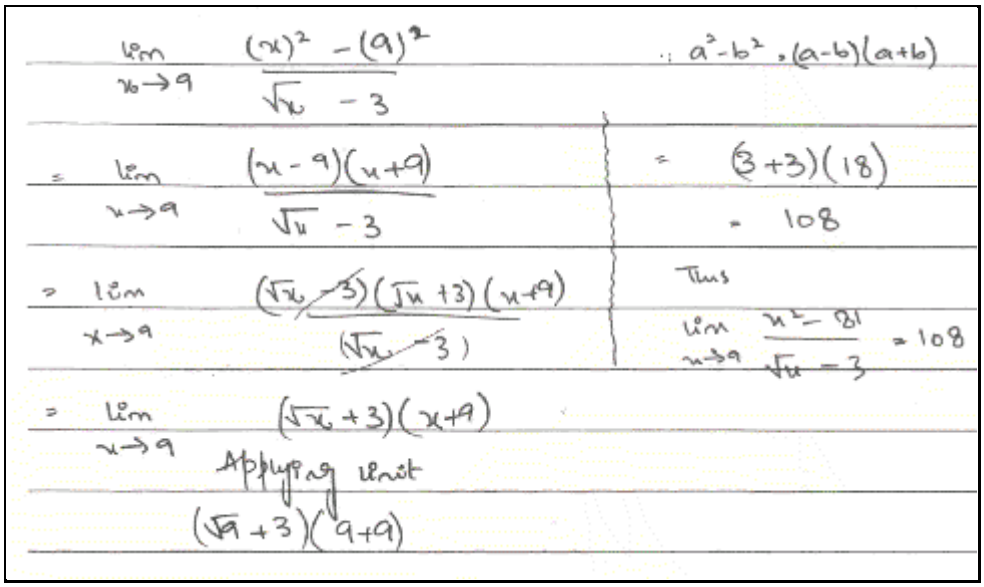
This was a well attempted question.

Question 1i:

Evaluate $\lim_{x\to 9}\dfrac{x^2-81}{\sqrt{x}-3}$.

Better responses showed two kinds of methods of solution. Candidates either used rationalisation or factorised $(x-9)$ into $\left(\sqrt{x}+3\right)\left(\sqrt{x}-3\right)$. In both cases, the limit was applied correctly and the answer was found.

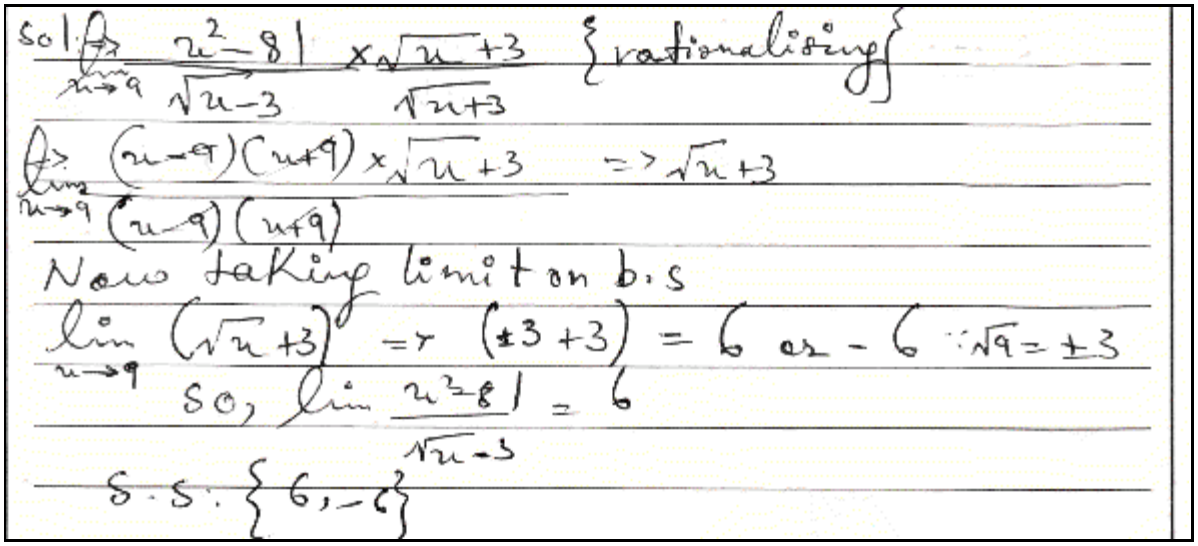
Example:



Weaker responses reflected weak understanding of limit of a function. The most frequent errors made by candidates were that they

i. directly applied limit to the given fraction without rationalising it.
ii. considered the denominator $\sqrt{x}-3$ as $\sqrt{x-3}$.
iii. incorrectly expanded x^2-81.

Example:



Question 1ii:

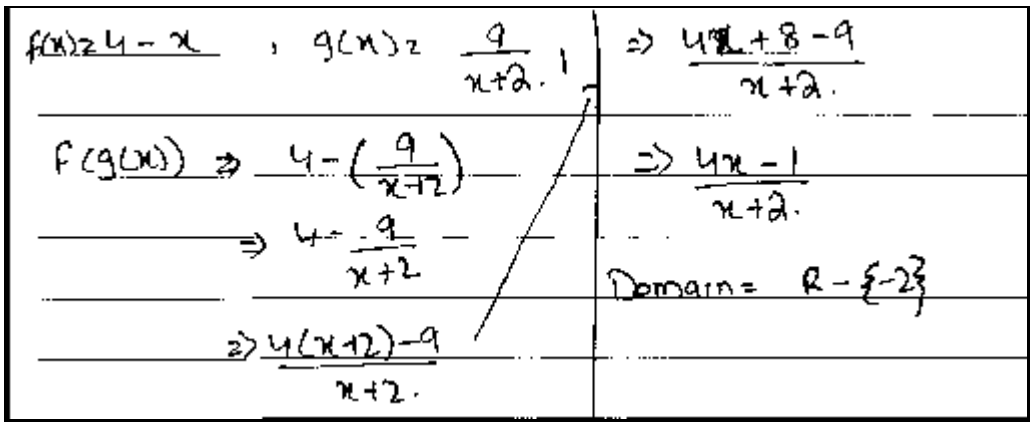
This was a well attempted question.

Question 1ii:

If $f(x)=4-x$ and $g(x)=\dfrac{9}{x+2}$, then find $f(g(x))$ and its domain.

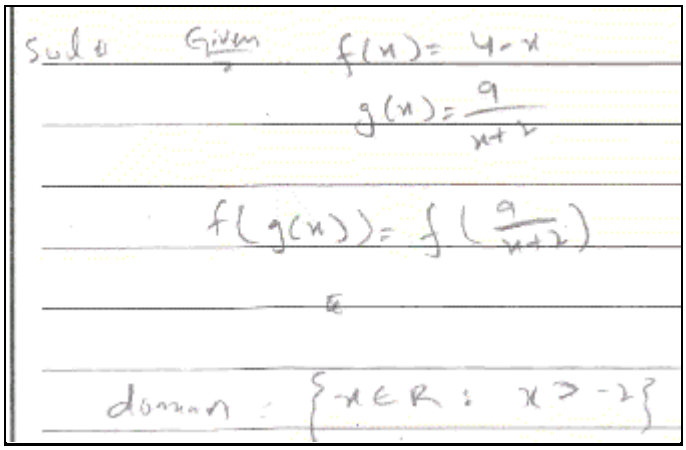
Better responses exhibited correct understanding of composite function. The function $g(x)$ was substituted in $f(x)$ to find $f\,g(x)$. The domain was also found easily by the candidates.

Example:



Weaker responses reflected that while most of the candidates were able to find the required composite function, they were unable to find the domain.

Example:



Question 2:

This question offered a choice to attempt any two questions from part a, b and c. Majority of the candidates attempted part b and c as compared to part a. However, part a and b were attempted better than part c.

Question 2a:

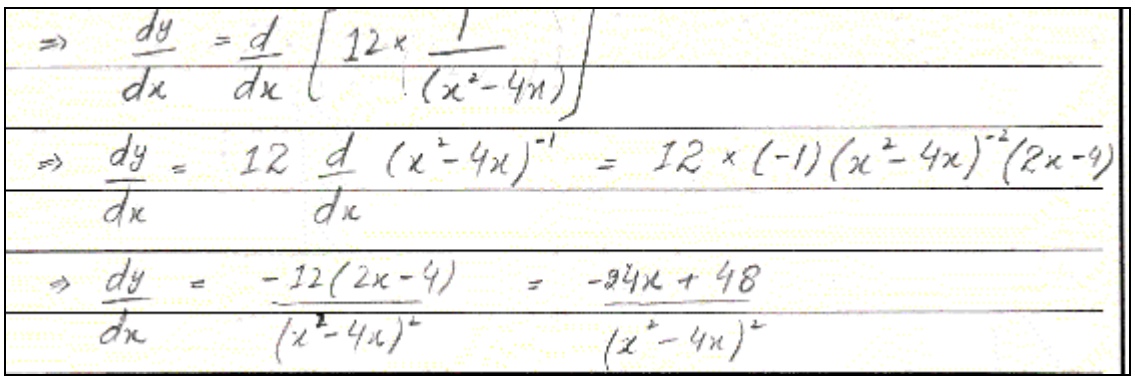
A curve has the equation $y = \frac{12}{x^2 - 4x}$. Find

i. $\frac{dy}{dx}$.

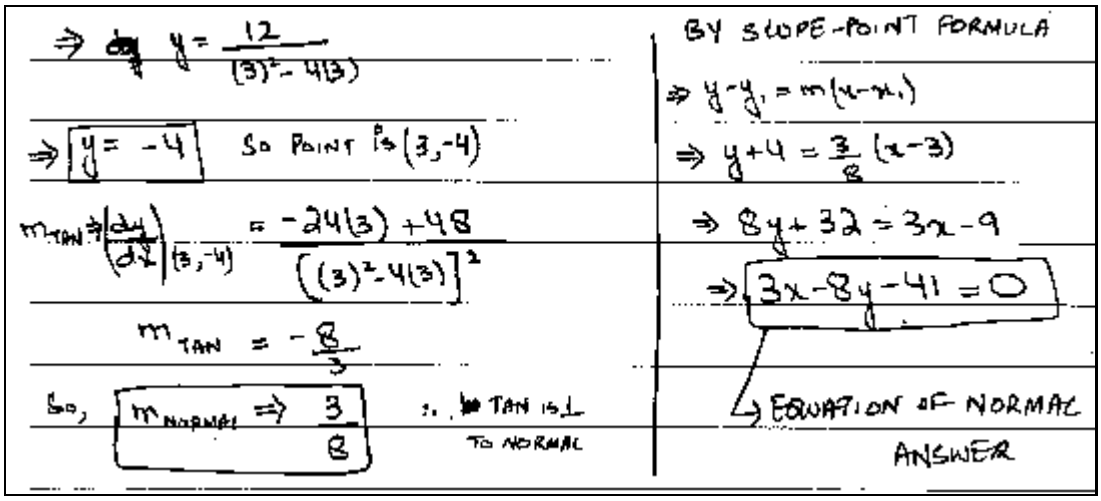
ii. the equation of the normal to the curve y at $x = 3$.

Better responses showed the correct differentiation of algebraic functions. Some candidates used product rule while others used quotient rule. The candidates were able to exhibit the process of finding equation of normal to a curve at a given point.

Example:

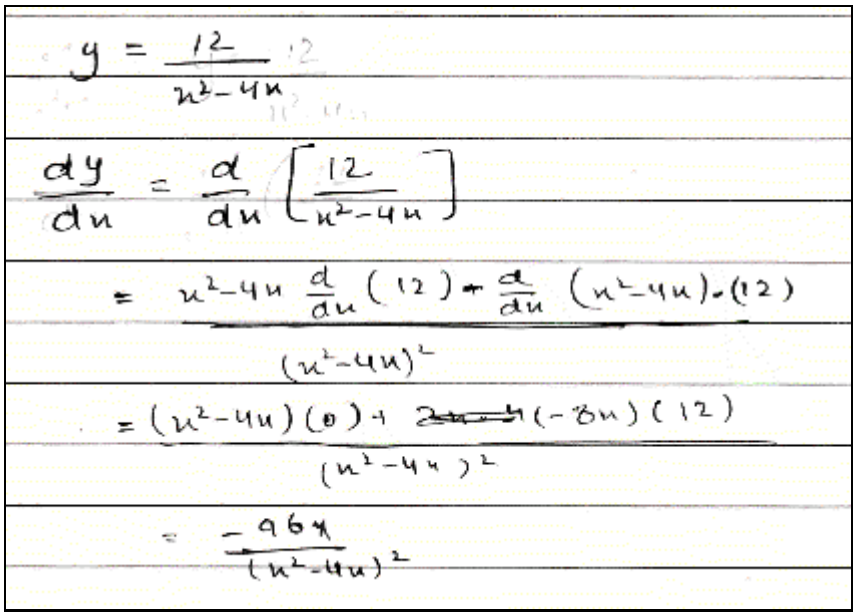


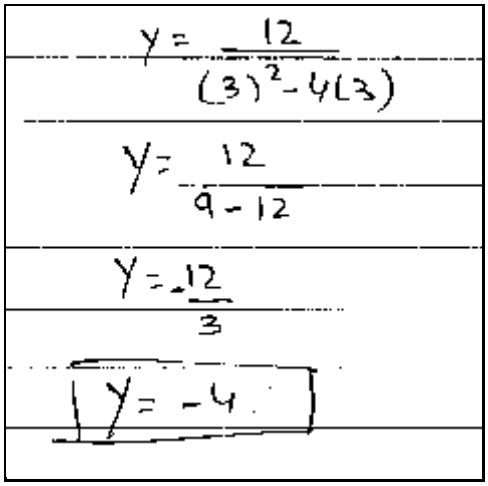
Example:



Weaker responses showed that, while most of the candidates were able to find $\left.\frac{dy}{dx}\right|_{x=3} = -\frac{8}{3}$, they did not know how to continue from there to find the equation of normal to the curve at the given point. Many candidates did not find the *y*-coordinate of the curve at $x = 3$. Those who did, substituted $x = 3$ in $\frac{dy}{dx}$ instead of *y*. Using the slope – 8/3, instead of 3/8, in the equation of normal was also a frequent mistake. Some candidates tried to find the equation of normal using two points form instead of point slope form which could not lead them to the answer. Another frequent mistake observed was the incorrect derivative of numerator, i.e. 12 found to be equal to 12 instead of 0.

Example:





Question 2b:

Find the derivative of the following with respect to *x*.

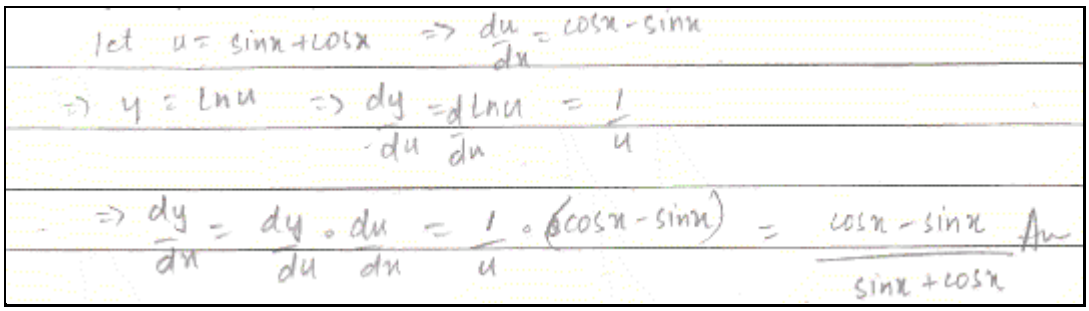
i. $y = \ln(\sin x + \cos x)$

ii. $y = \dfrac{x + e^{3x}}{\sqrt{x}}$

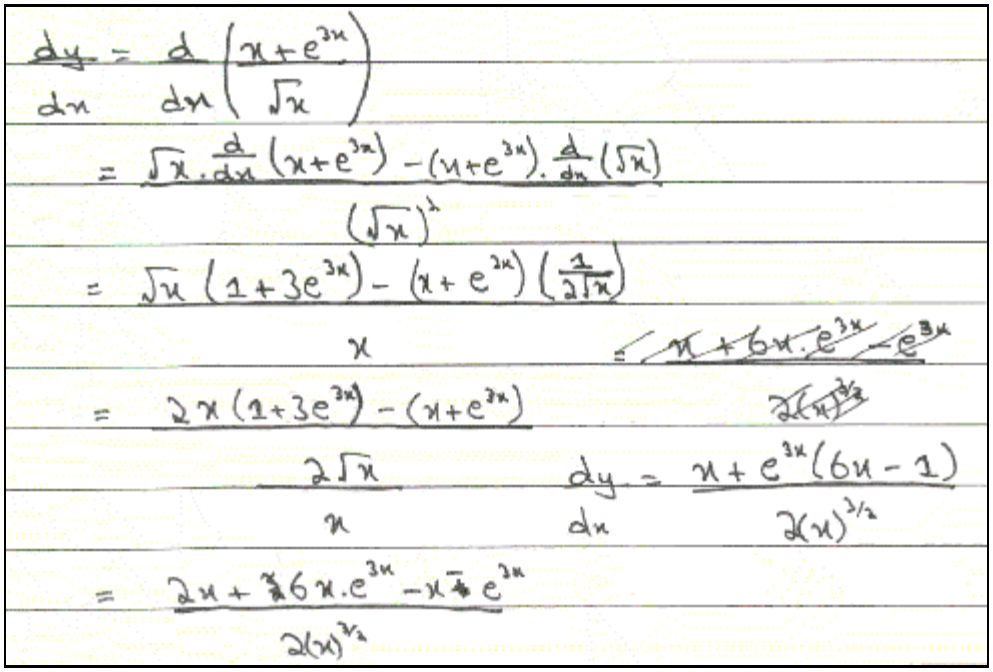
Better responses exhibited correct application of derivative of logarithmic functions in part i and quotient of two functions in part ii. The differentiation of composite functions was also found correctly in both parts. The following multiple methods were observed in part ii:

i. The given function was first expressed as $y = \left(x + e^{3x}\right)\left(\sqrt{x}\right)^{-1}$ and then derivative of product of two functions was found.
ii. The given function was first rationalised and then derivated.

Example:

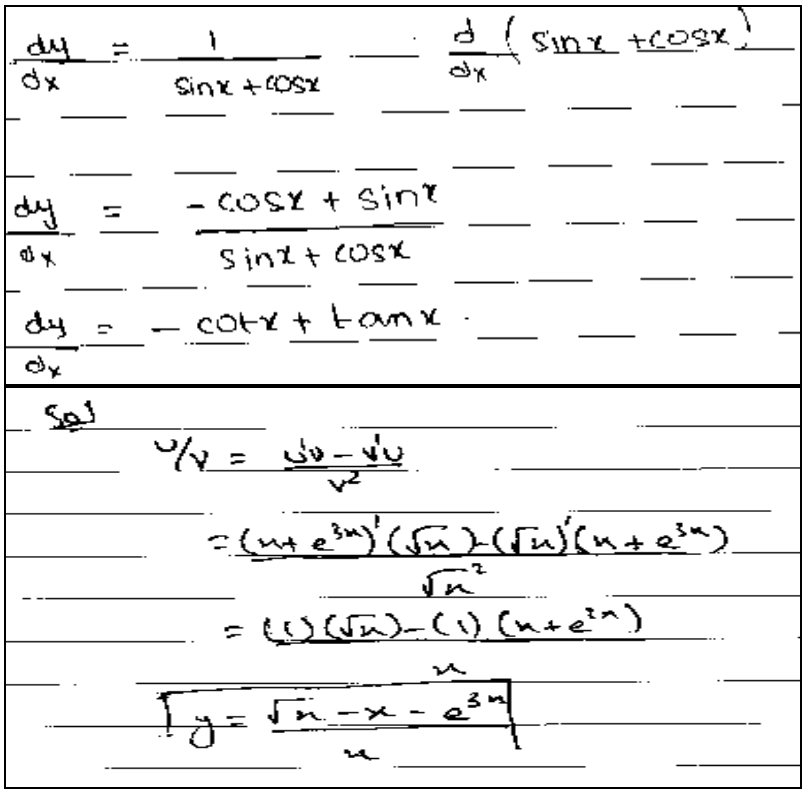


Another example:



Weaker responses reflected weak concepts in application of differentiation on functions. In part i, candidates expanded the given function as $y = \ln \sin x + \ln \cos x$ and then took derivative of each term. In part ii, candidates made mistakes in the application of differentiation of quotient of two functions. The most common mistake was the use of positive sign instead of negative.

Example:

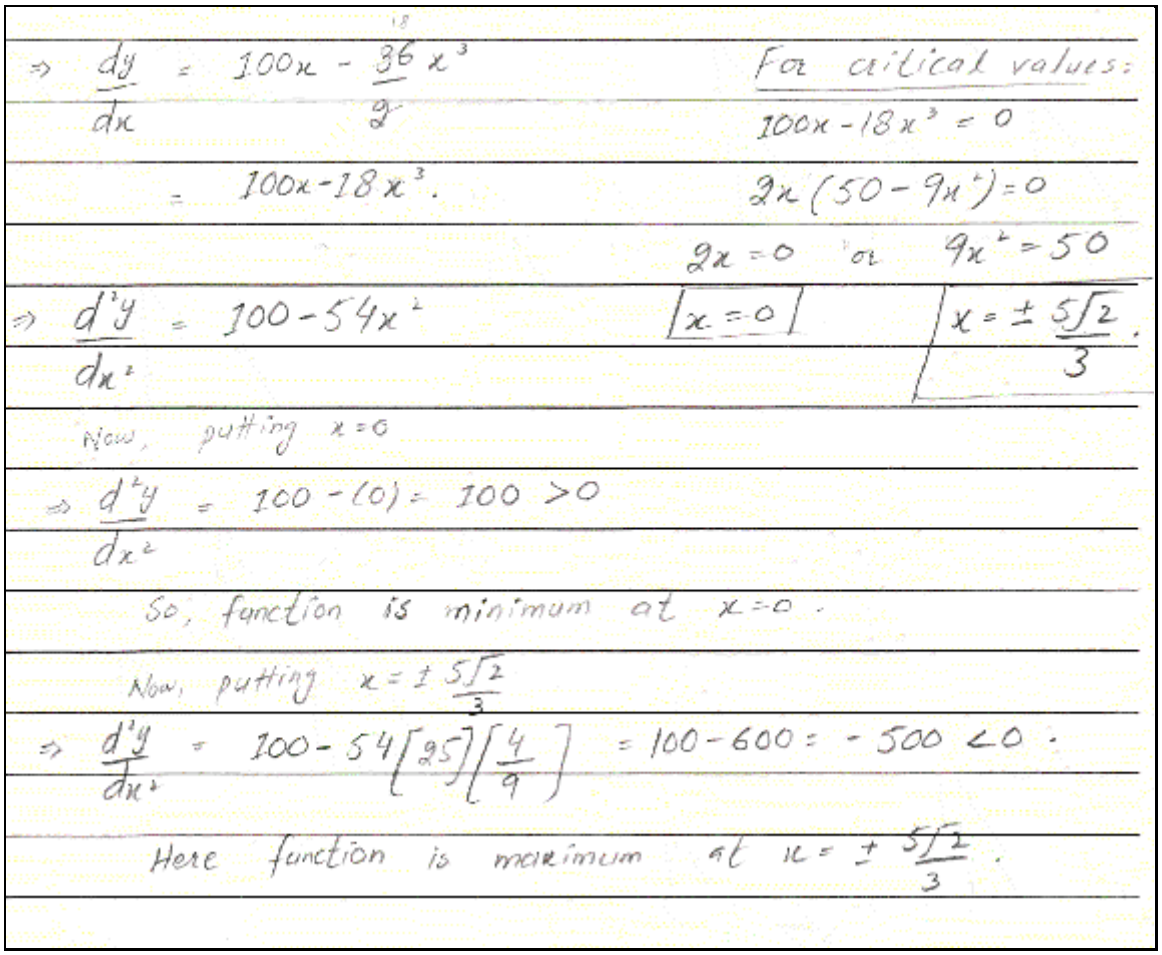


Question 2c:

For the function $y = 50x^2 - \frac{9}{2}x^4$, find the values of x for which the function y has extreme values. Apply second derivative rule to find the minimum value of the given function.

Better responses show the correct application of differentiation to find the extreme values of a function. The first and second derivative tests were used to find extreme points and their nature. Hence, the minimum value of the function was found.

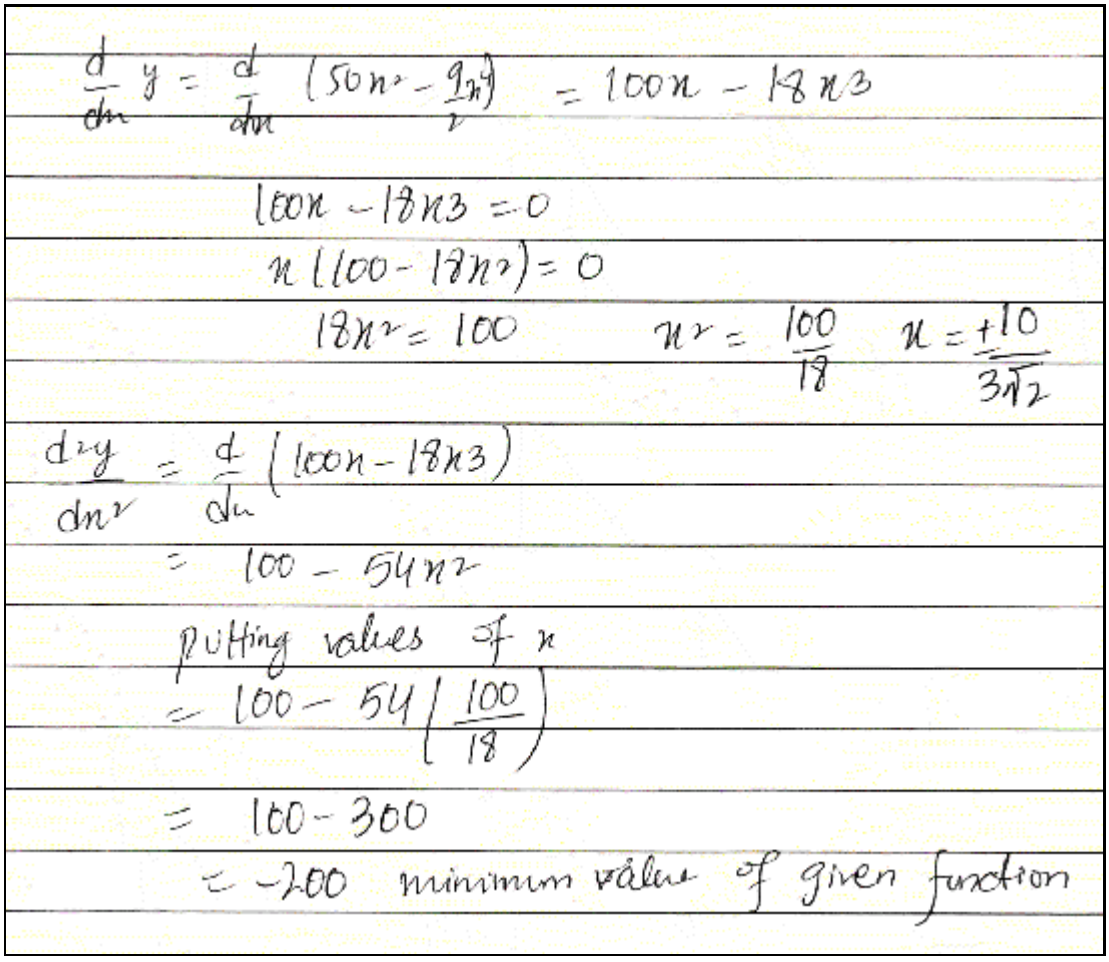
Example:



Weaker responses reflected the confusion of candidates in second derivative test. Most of the candidates were able to find the first derivative, put it equal to zero, find the extreme values, and find second derivative. After that, the following mistakes were common:

i. The maximum point was taken as minimum and minimum point taken as maximum.
ii. The nature of points was not found and the minimum value of the function was found directly.
iii. The nature of the point was found using second derivative test but the answer was left incomplete after that. The minimum value of the function was not found.

Example:



Question 3:

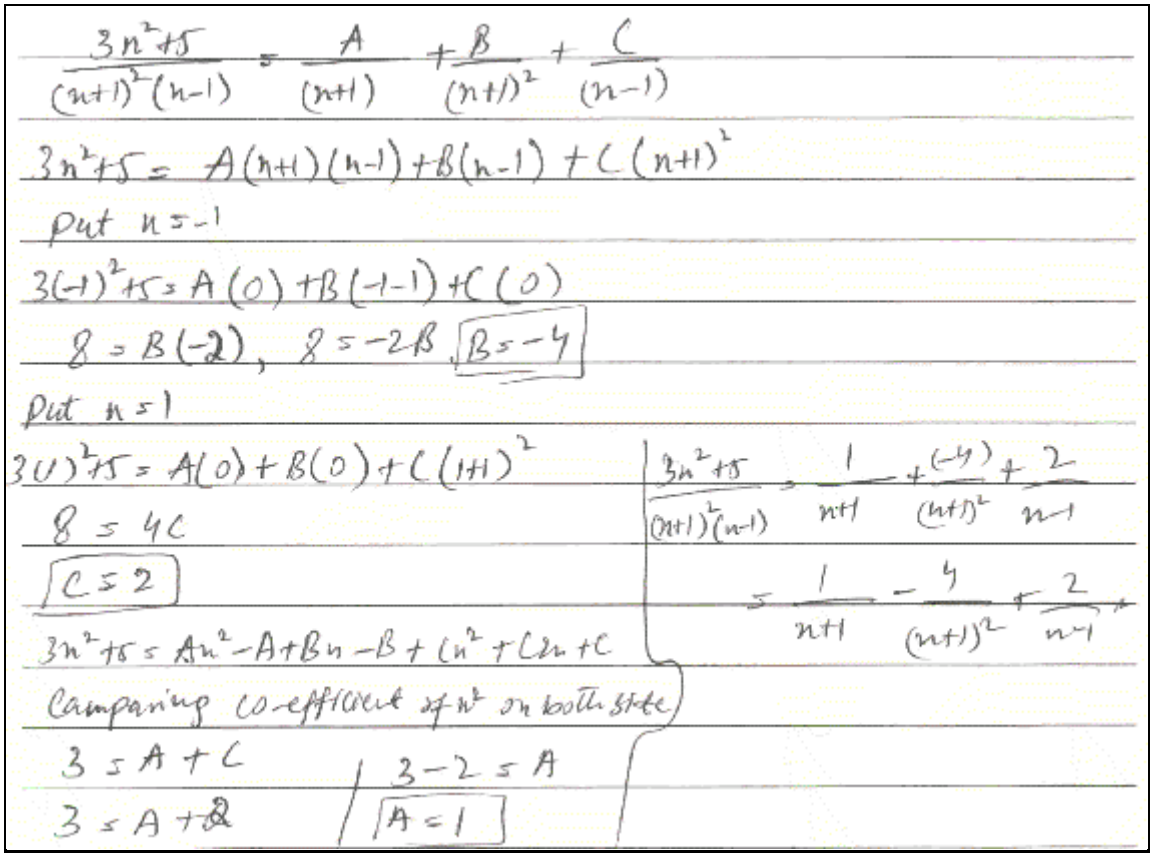
This question offered a choice to attempt any two questions from part a, b and c. Majority of the candidates attempted part a and c. All parts were well attempted.

Question 3a:

Resolve $\dfrac{3x^2+5}{(x+1)^2(x-1)}$ into its partial fractions.

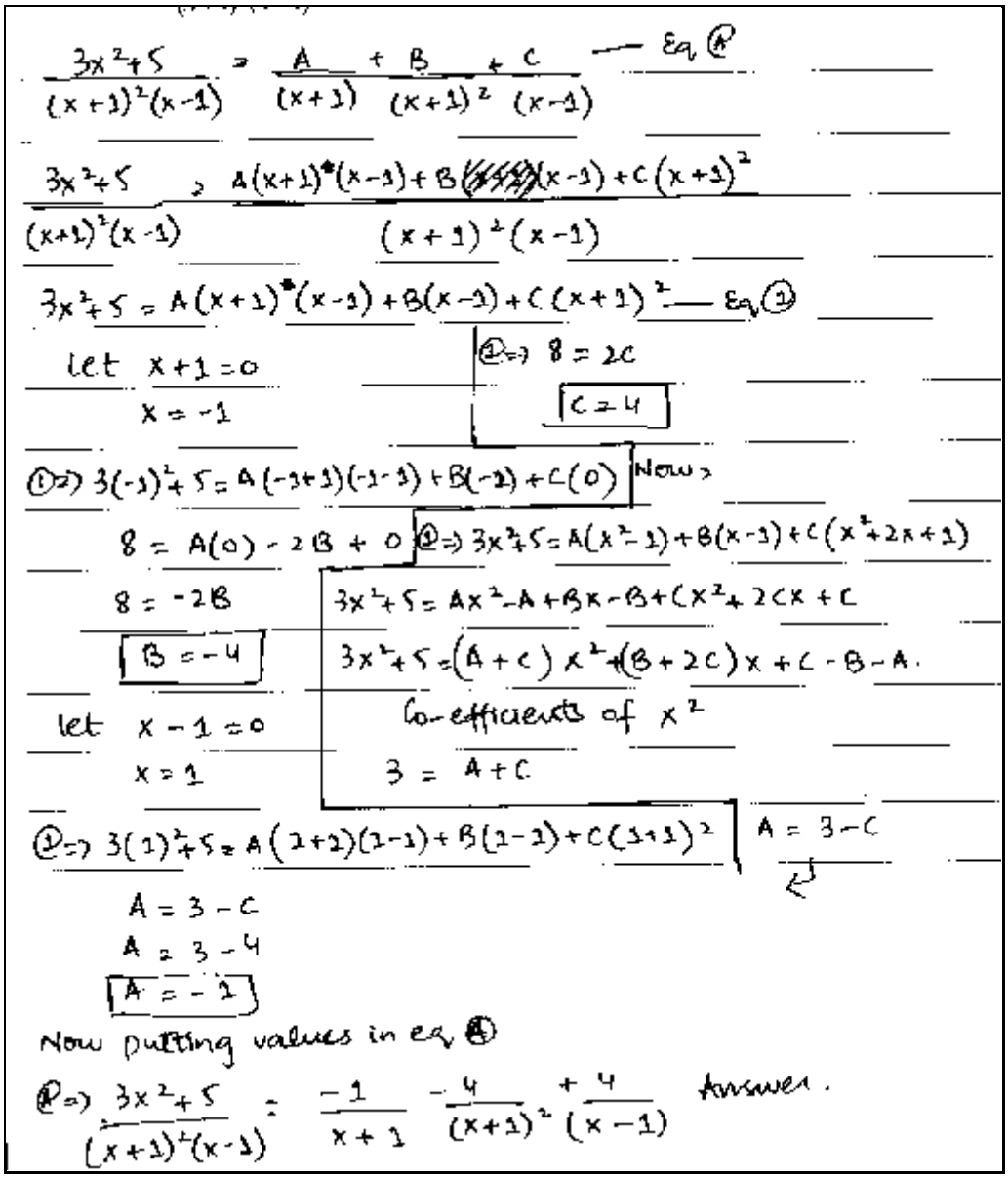
Better responses converted the given function into the correct form, i.e. $\dfrac{3x^2+5}{(x+1)^2(x-1)}=\dfrac{A}{x+1}+\dfrac{B}{(x+1)^2}+\dfrac{C}{x-1}$, eliminated the denominator and found the values of *A*, *B* and *C*. After substitution of *A*, *B* and *C* in the fraction, the answer was found.

Example:



Weaker responses reflected that candidates had difficulty expressing the given fraction in terms of partial fractions due to the repeated root $(x+1)^2$. Hence, the most frequent incorrect responses were $\frac{3x^2+5}{(x+1)^2(x-1)} = \frac{A}{(x+1)^2} + \frac{B}{x-1}$ and $\frac{3x^2+5}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{Bx+C}{(x+1)^2} + \frac{D}{x-1}$.

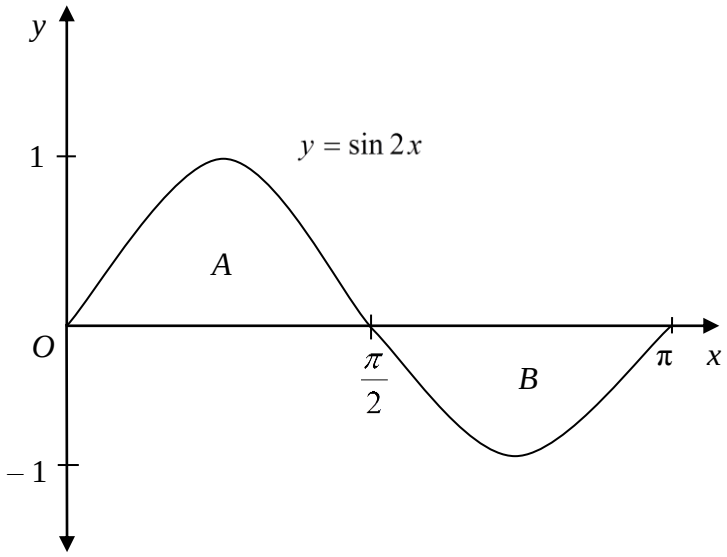
Example:



Question 3b:

The given figure shows the region *A* and *B* enclosed by the curve $y = \sin 2x$ and *x*-axis.

i. Find the area of region *A*.
ii. Hence, find the area of region *B*.



Better responses showed that in part i, candidates integrated the equation of curve as $\int_0^{\frac{\pi}{2}} \sin 2x\, dx$ to find the area of region *A*. Despite use of the word 'hence', most of the candidates could not connect part i with ii to infer that region *A* is symmetric to region *B* so they have the same areas. Such candidates found the integral $\int_{\frac{\pi}{2}}^{\pi} \sin 2x\, dx$ which was correct but time consuming.

Example:

i. Find the area of region A. (5 Marks)

$y = \sin 2x$

$$\text{Area of } A = \int_0^{\pi/2} \sin 2x \, dx$$

$$= \left[-\frac{\cos 2x}{2} \right]_0^{\pi/2}$$

$$= \left[-\frac{\cos 2(\pi/2)}{2} \right] - \left[-\frac{\cos 0}{2} \right]$$

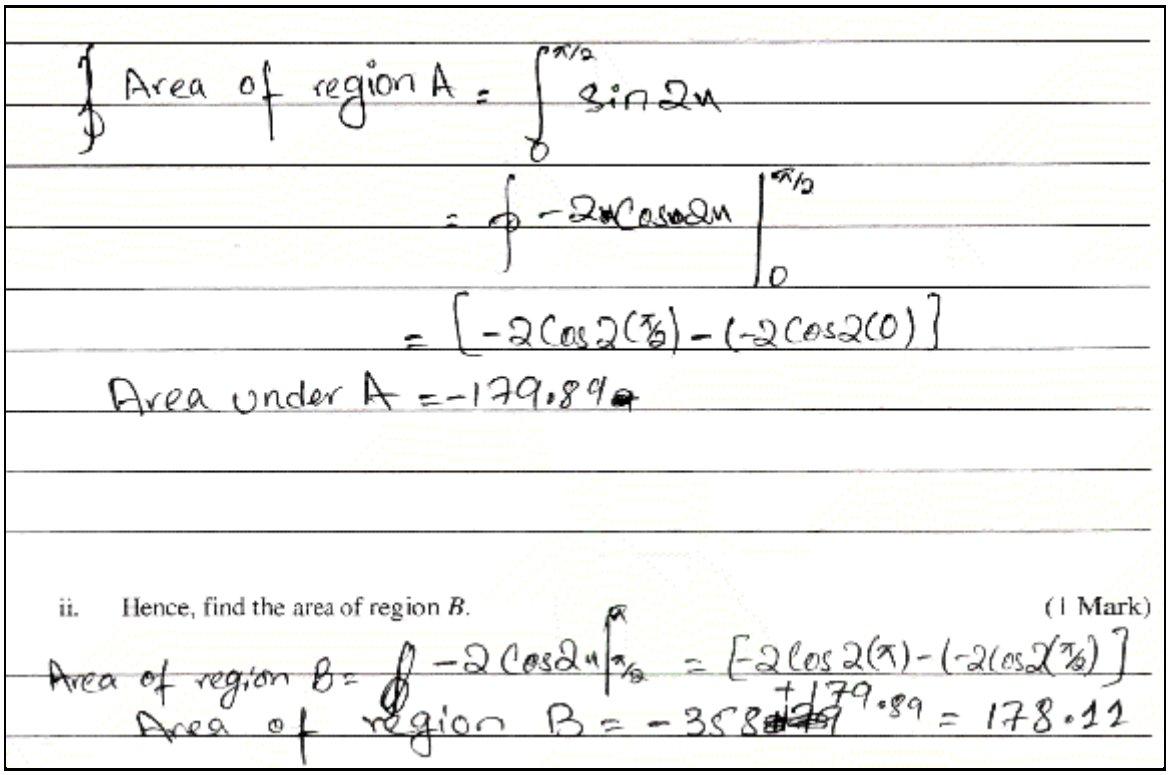
$$= \left[\frac{1}{2} + \frac{1}{2} \right] \Rightarrow \frac{2}{2} \Rightarrow \text{Area of region A} = 1 \text{ unit}$$

ii. Hence, find the area of region B. (1 Mark)

Area of region B = 1 unit

Weaker responses showed that in part i, candidate were mostly able to express the correct integral required to find the area of region A, i.e. $\int_0^{\frac{\pi}{2}} \sin 2x\, dx$ but they made errors in integration of sin2*x*. Many candidates did not treat it as a composite function and found the integral as $-\cos 2x$ instead of $\frac{(-\cos 2x)}{2}$. The negative sign was missed out by many students too.

Example:

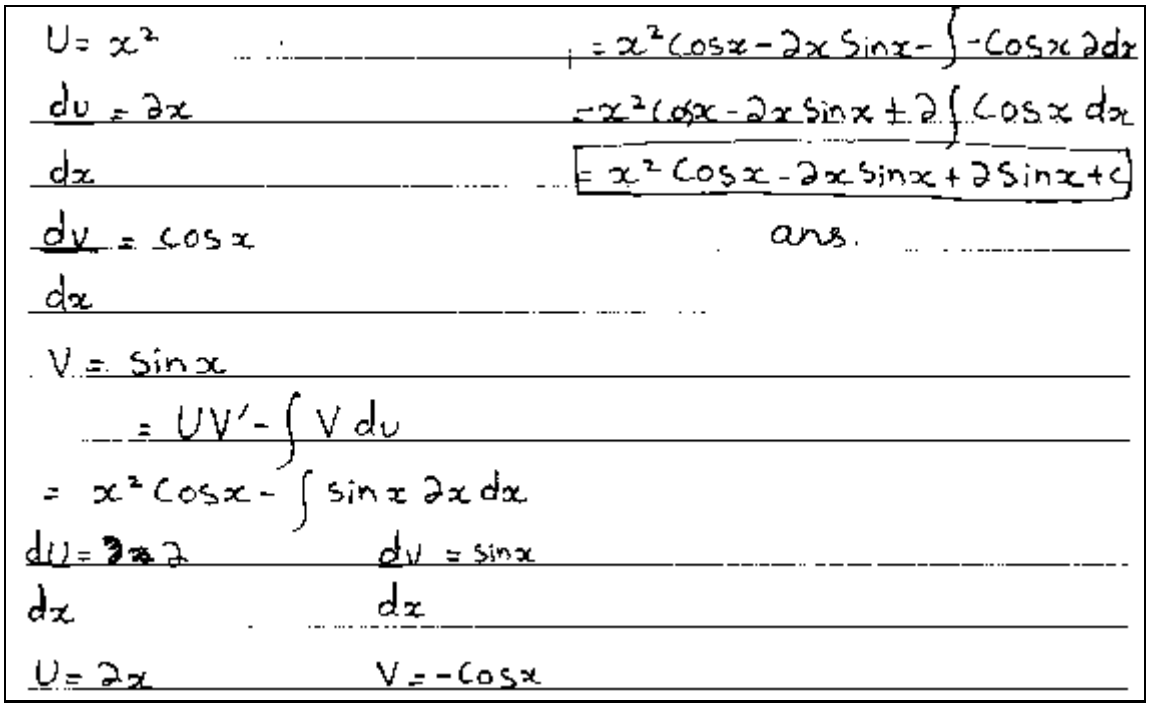


Question 3c:

Evaluate $\int x^2 \cos x\, dx$.

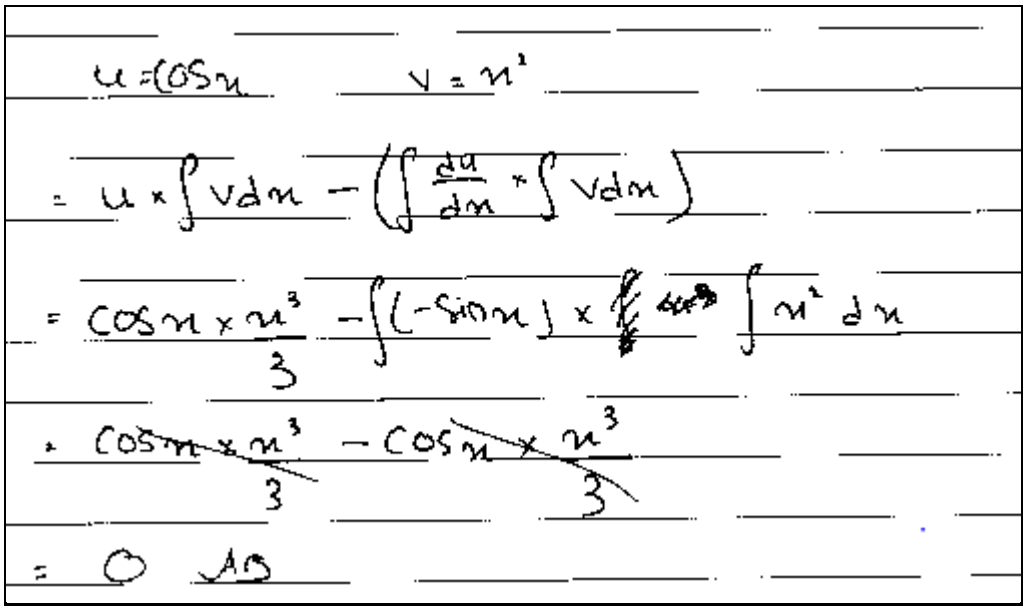
Better responses indicated that most of the candidates were able to apply integration by parts. They were able to choose $f(x) = x^2$ and $g'(x) = \cos x dx$ and substitute the values in the formula of integration by parts $\int x^2 \cos x dx = x^2 \sin x - 2\int x \sin x dx$. The second term was again integrated by parts to get $\int x^2 \cos x dx = x^2 \sin x - 2\left(-x\cos x + \int \cos x dx\right)$ and hence the final answer was found.

Example:



Weaker responses reflected that in part i, candidates chose $f(x)=\cos x$ and $g'(x)=x^2dx$, i.e. the opposite of what was required. Due to this selection, the process of integration by parts grew more difficult as they proceeded with it and eventually gave up.

Example:



Question 4:

This question offered a choice to attempt any two questions from part a, b and c. Majority of the candidates attempted part a and b. Part a and b were well attempted but part c was not well attempted.

Question 4a:

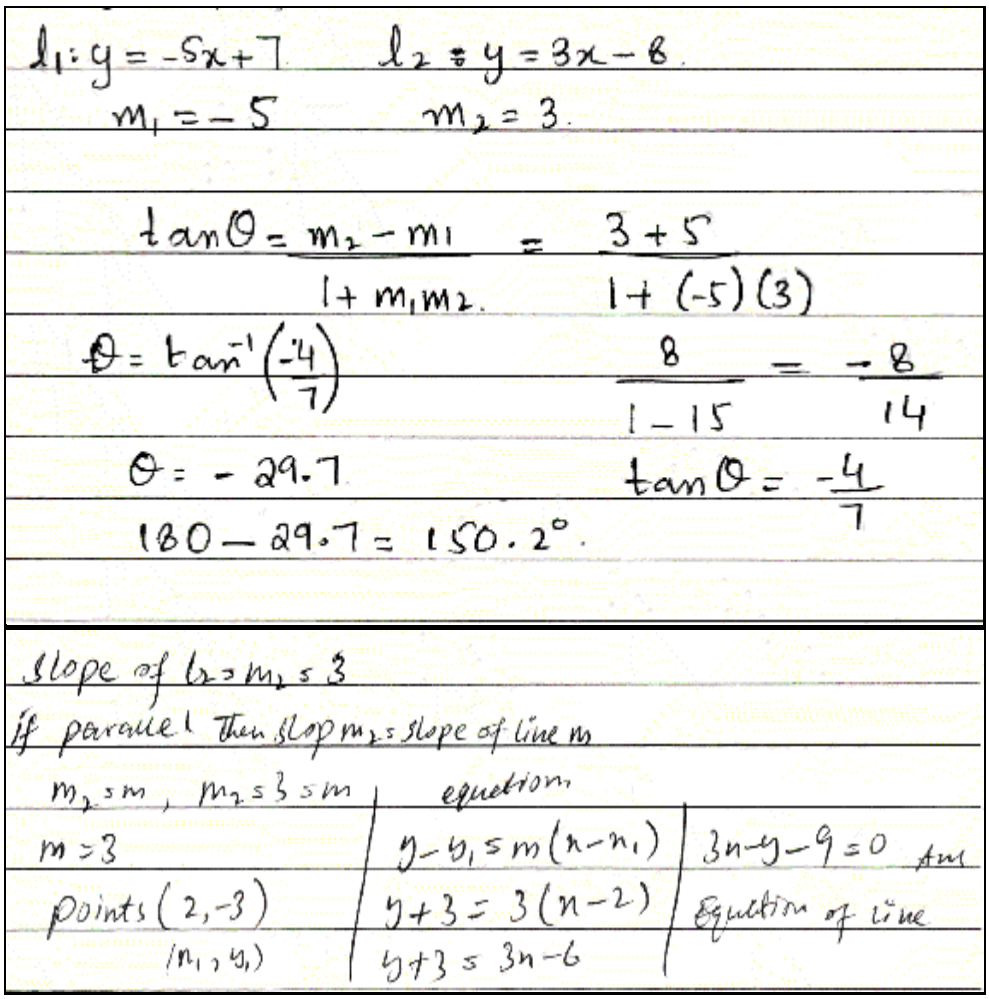
For the lines $l_1: y=-5x+7$ and $l_2: y=3x-8$, find the

i. angle from l_1 to l_2.
ii. equation of a line that is parallel to l_2 and passes through the point $(2,-3)$.

Better responses of part i reflected correct application of formula and substitution of values m_1 and m_2. The angle from l_1 to l_2 was found, i.e. 150.26°, by finding basic angle using $\tan\theta = -\frac{4}{7}$. Majority of the candidates, however, did not find the basic angle. Therefore, the most frequent answer found by candidates was – 29.74° which is the clockwise angle instead of the anticlockwise answer required. Due to high frequency of such responses, it was decided that such responses will be given full credit.

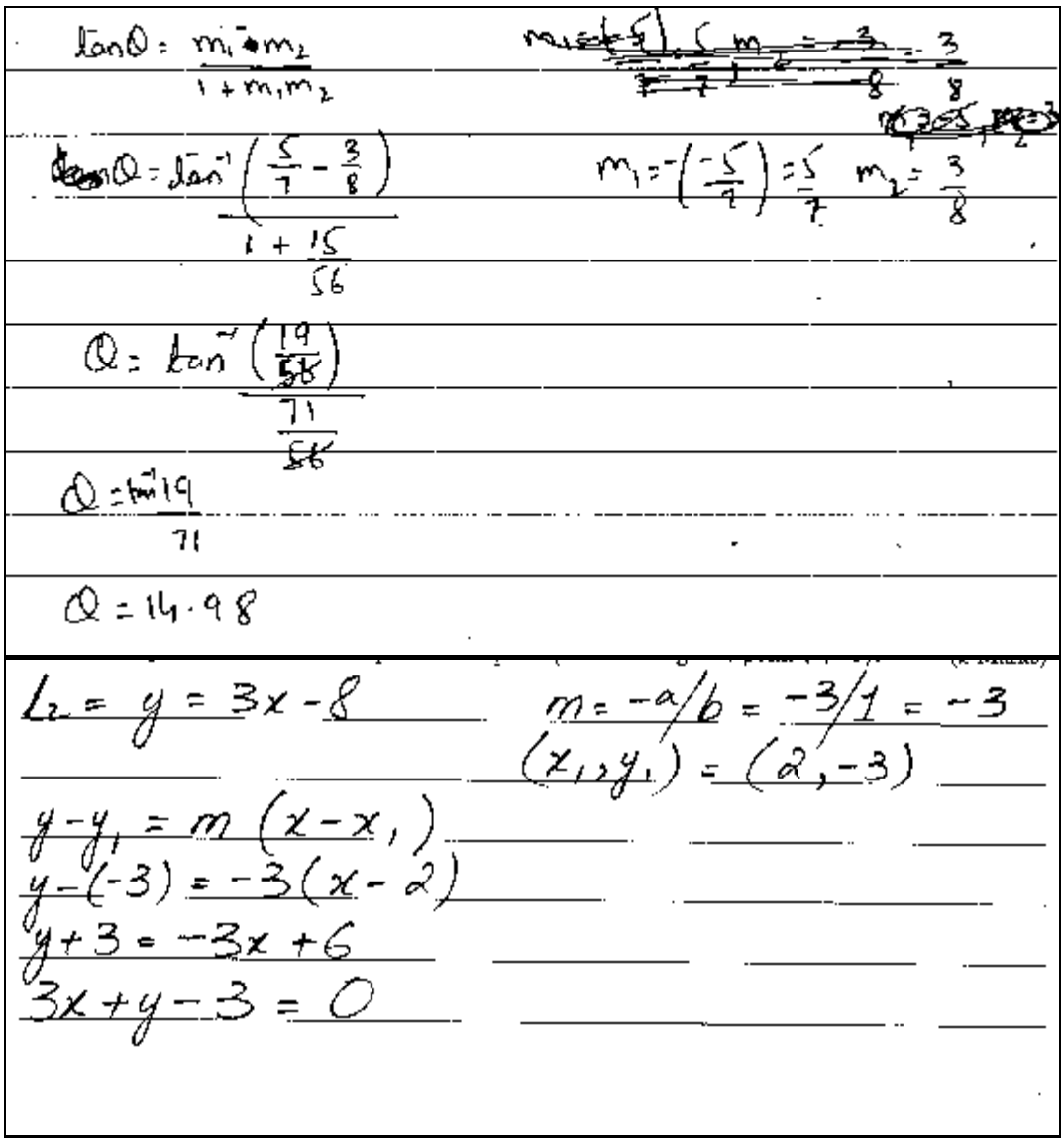
In part ii, the concept that parallel lines have same slope was reflected. The given point was used in point slope form to find the equation.

Example:



Weaker responses of part i exhibited incorrect substitution of m_1 and m_2 in the formulae. After substitution, minor arithmetic errors were frequent too. Part ii reflected candidates' confusion in slope of parallel lines since they used slope of perpendicular lines instead of parallel. Minor algebraic simplification errors were also frequent.

Example:



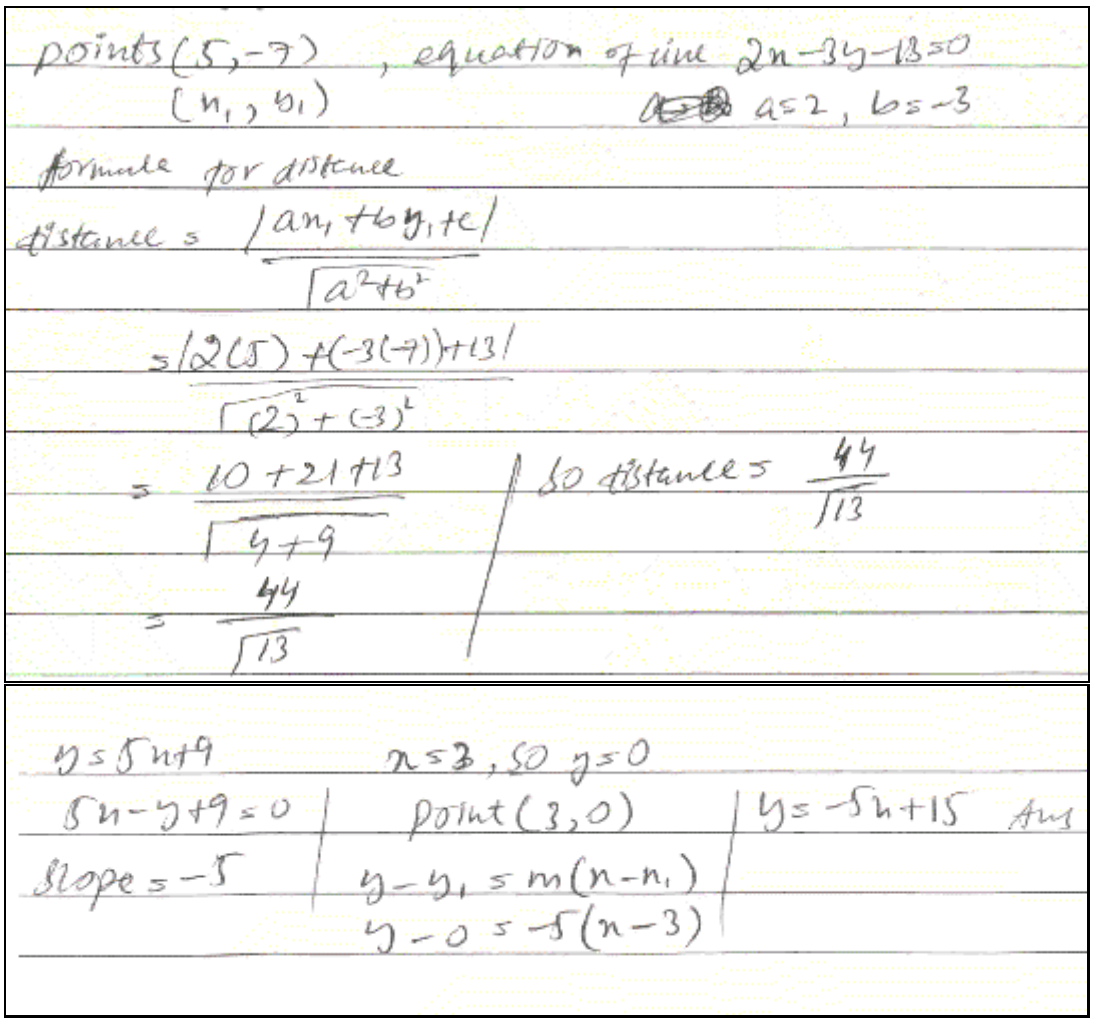
Question 4b:

i. Find the perpendicular distance of a point (5, – 7) to the line $2x-3y-13=0$.

ii. Find the equation of a line that is perpendicular to the line $y=5x+9$ and cuts *y*-axis at 3.

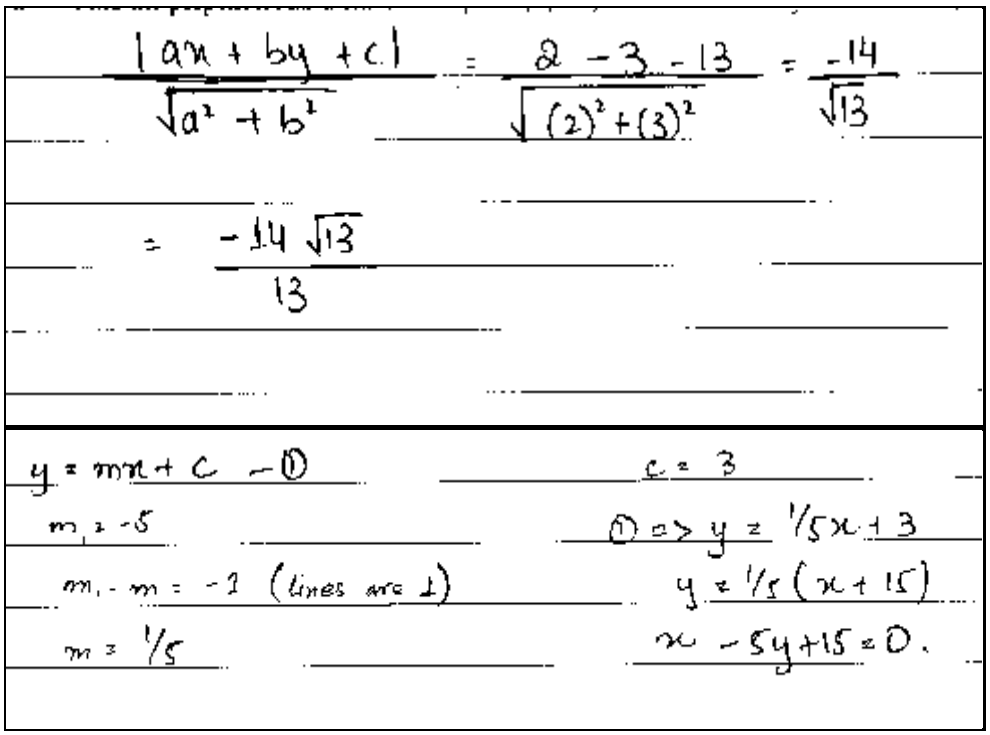
Better responses exhibited correct substitution in the formula of distance of point from a line $d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}}$. After simplification, the candidates found the correct answer. In part ii, the concept of slope of perpendicular lines as reciprocals was reflected. The given *y*-intercept was used to find the point (0, 3) which was substituted in point-slope form to find the equation.

Example:



Weaker responses of part i reflected that while most of the candidates used the correct formula $d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$, they did not know what a, b, c, x_1, y_1 stand for. Due to this, the most common error that was observed in this question was incorrect denominator, i.e. $\sqrt{5^2 + (-7)^2}$. In part ii, candidates made mistake in finding the point at *y*-intercept and slope of perpendicular line. Minor errors in algebraic simplification were also frequent.

Example:

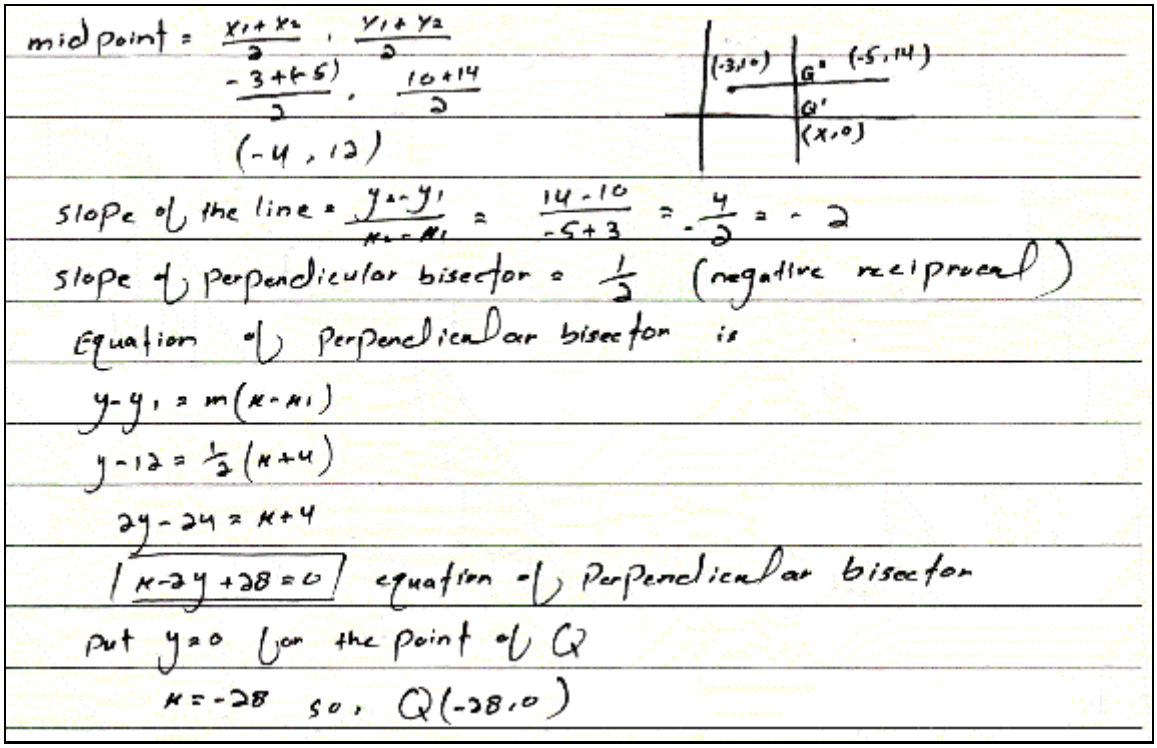


Question 4c:

Find equation of perpendicular bisector of the line joining the points (– 3, 10) and (– 5, 14). This perpendicular bisector meets *x*-axis at a point *Q*. Find the co-ordinates of the point *Q*.

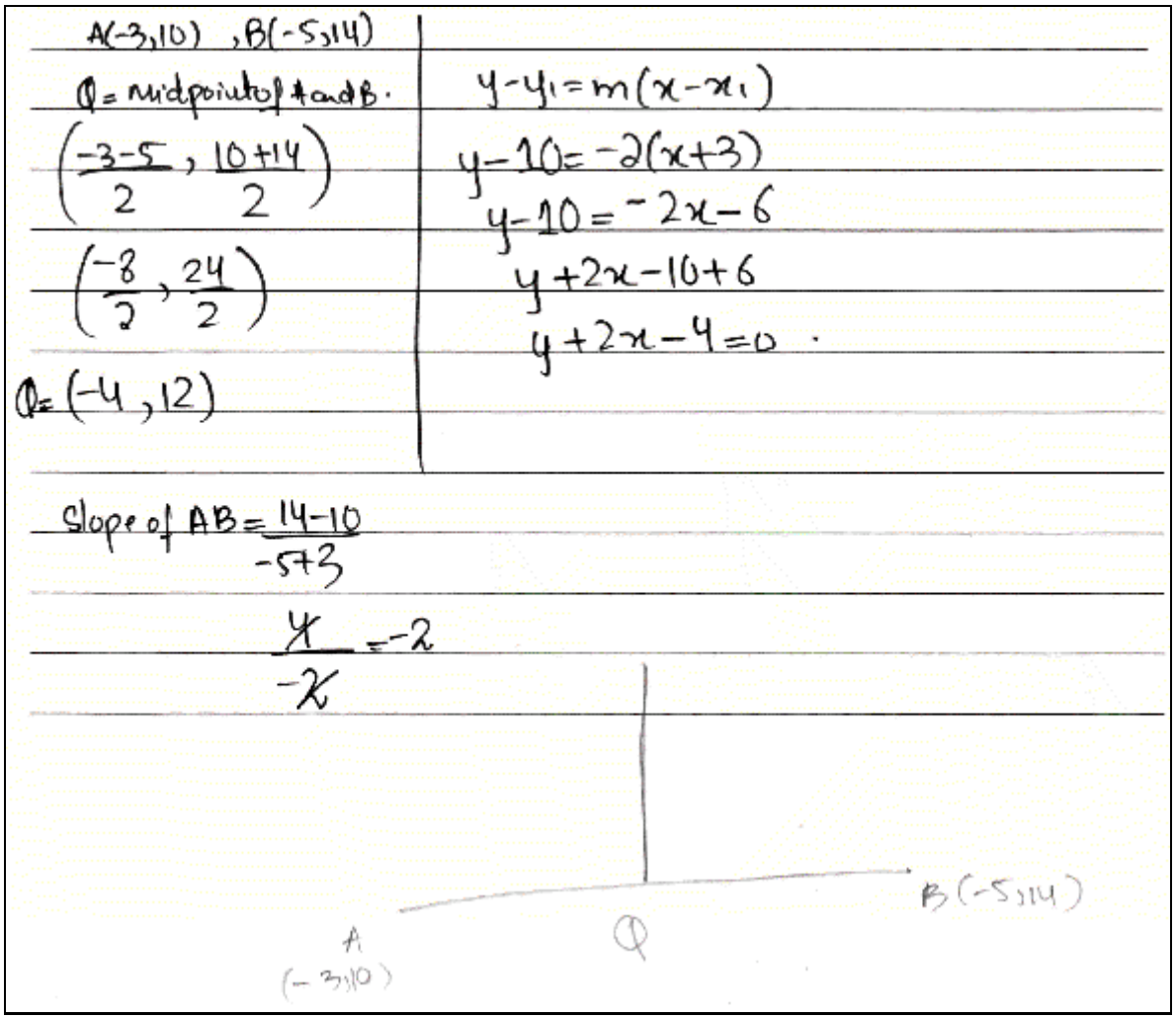
Better responses exhibited that candidates were able to visualise the given data and get the solution. Many candidates sketched the perpendicular bisector and point *Q* on *xy* coordinate system (The points (− 3, 10) and (− 5, 14) do not lie on a straight line, but it does not matter since it is just a rough sketch drawn by candidates for themselves). The midpoint of the given points and their gradient was thus found which was used to find the equation of perpendicular bisector. Substituting $y = 0$ in that equation provided the value of *Q*.

Example:



Weaker responses showed that many candidates carelessly attempted the question and did not complete it. The value of Q was not found after finding the equation of perpendicular bisector. When finding the equation of perpendicular bisector, the most frequent mistake was using the points $(-3, 10)$ and $(-5, 14)$ instead of the midpoint $(-4, 12)$ between these two points.

Example:



Question 5:

This question showed average performance.

Question 5:

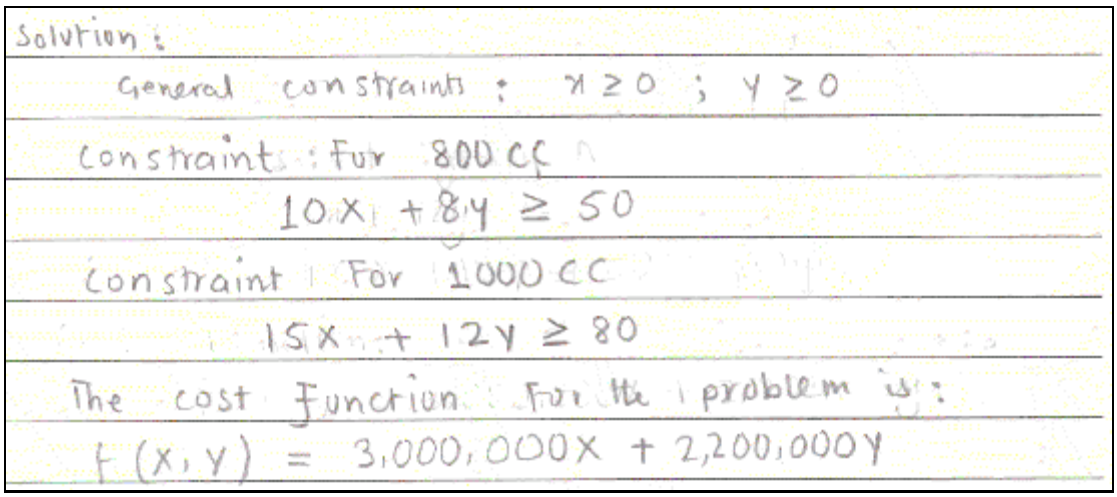
A car manufacturing company manufactures 800 CC and 1000 CC cars in two factories *A* and *B*. The number of cars manufactured daily in each factory is shown in the given table.

	800 CC	**1000 CC**
Factory *A*	10	15
Factory *B*	8	12

The daily cost of operating the factories *A* and *B* is Rs 3,000,000 and Rs 2,200,000 respectively. Factories *A* and *B* work together to complete an order of 50 cars of 800 CC and 80 cars of 1000 CC. If *x* and *y* are the number of days factories *A* and *B* took to complete the order, find the constraints and cost function to complete this order.

Better responses showed that the candidates were able to understand the given situation. The given variables *x* and *y* in the question were correctly used, along with the values given in the table, to construct meaningful constraints and cost functions.

Example:


Solution :

General constraints : $x \geq 0$; $y \geq 0$

Constraint : For 800 CC

$10x + 8y \geq 50$

Constraint : For 1000 CC

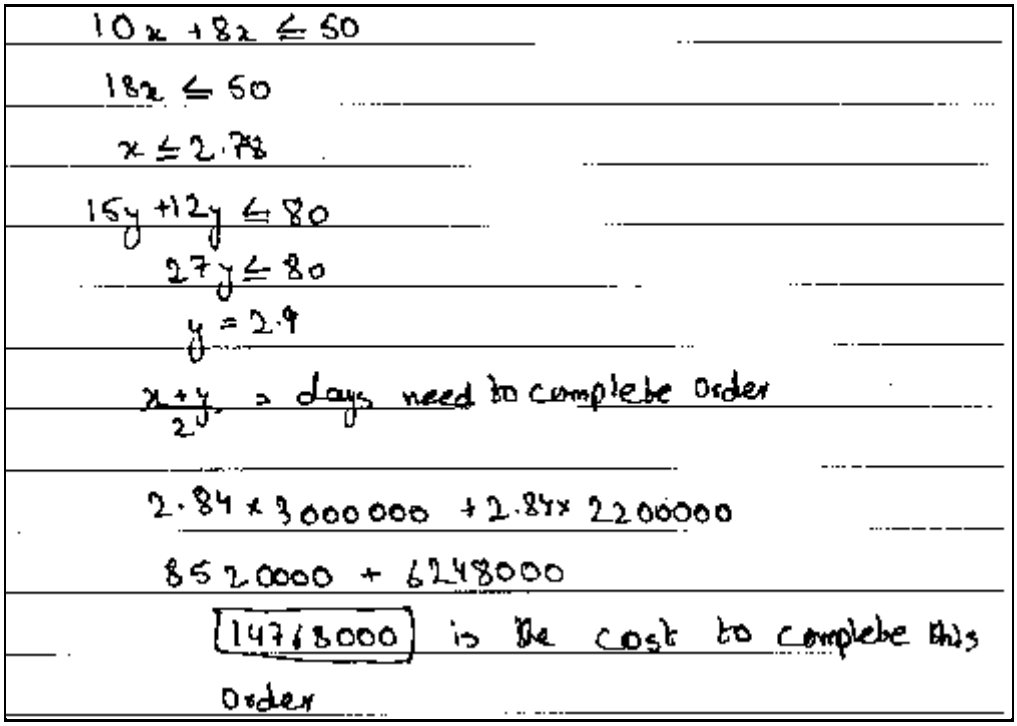
$15x + 12y \geq 80$

The cost function for the problem is :

$f(x, y) = 3{,}000{,}000x + 2{,}200{,}000y$

Weaker responses reflected weak understanding of application of linear programming with most of the students giving constrains and cost functions that made no mathematical sense according to the given situation. Since it is generally a difficult topic for candidates, to facilitate them, data was given in tabular form and *x* and *y* were already defined in the question. Even then, many candidates were not able to work with it.

Example:



Question 6:

This question offered a choice between part a and b. Majority of the students chose part a as compared to part b. Both parts showed average performance. Not many candidates were able to score full marks.

Question 6a:

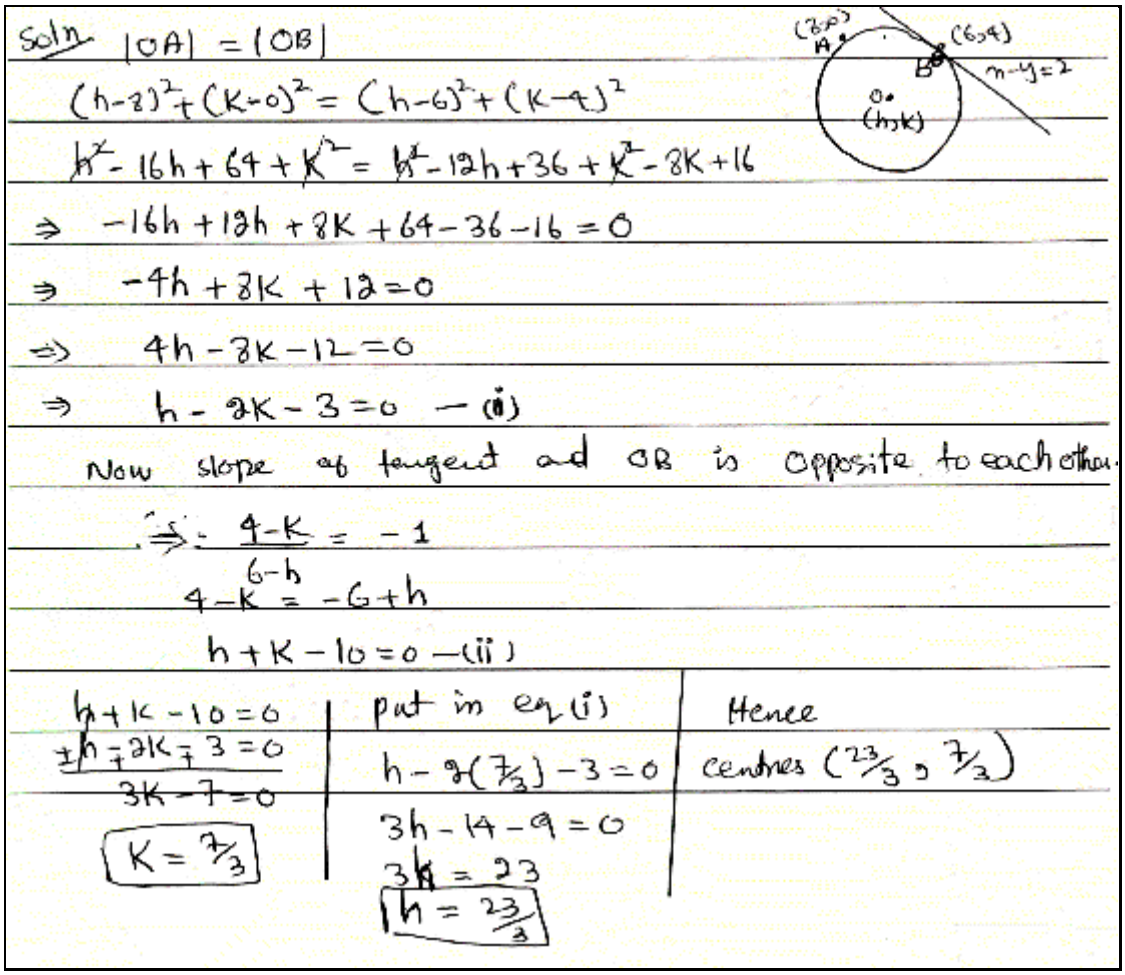
The line $x - y = 2$ is tangent to a circle at point (6, 4). The point (8, 0) lies on this circle. Find the centre of this circle.

Better responses exhibited multiple methods of solution. The most frequently used are as follows:

i. The given two points that lie on the circle were substituted in general form of equation of circle to find two equations in terms of three variables *f*, *g* and *c*. The gradient of radius was found as – 1 using the equation of tangent. The equation $\frac{4+f}{6+g} = -1$ was found which was used for simultaneous solution with the previous two equations to find the centre of the circle.

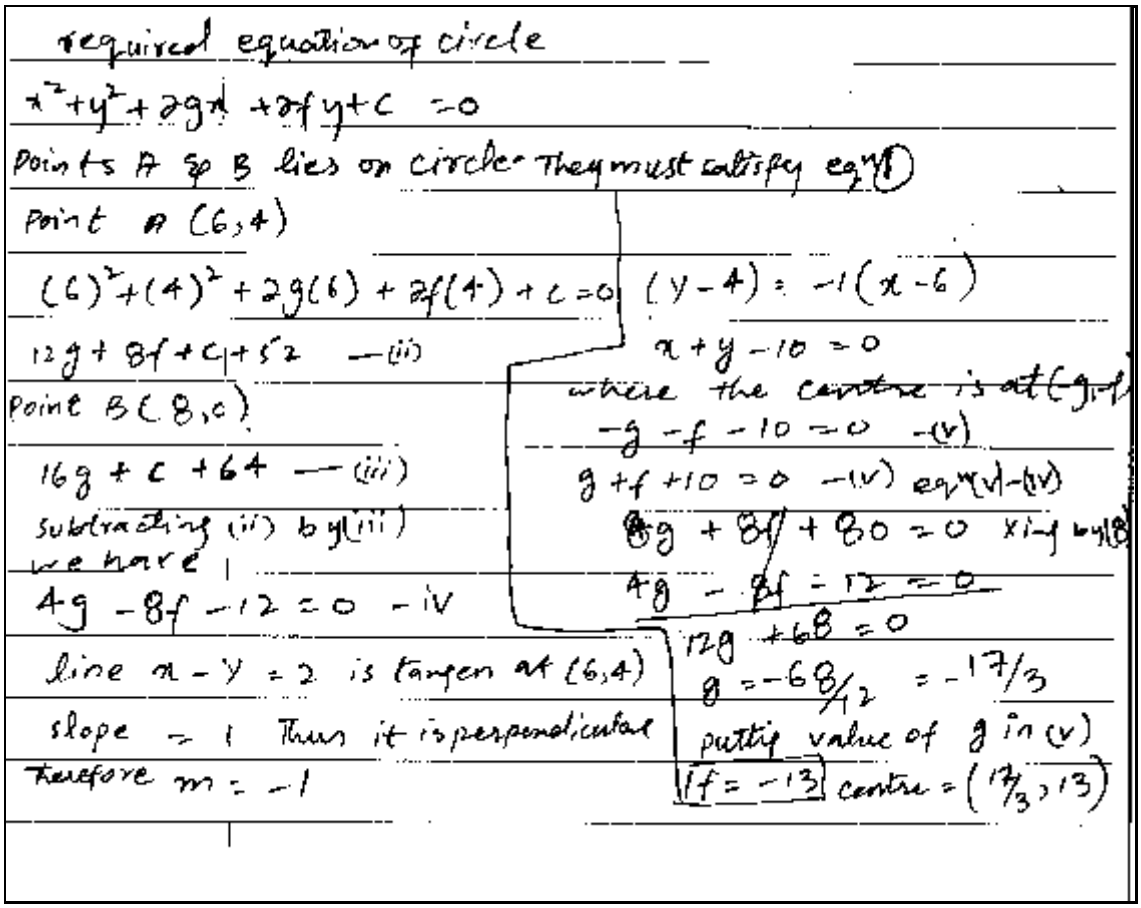
ii. The distance of the given two points from the centre was found in terms of *f* and *g* and was equated since both represent radius. The equation of line passing through the point (6, 4) and the centre of circle was found. These equations were solved simultaneously to find the answer.

Example:



Weaker responses showed that candidates were able to understand parts of question and not the whole. They could not visualize the situation and gauge how the given data can be used to find the centre of the circle. Most of the candidates substituted the given points to get equations but they did not know where to go from there. A lot of guess work was observed too.

Example:

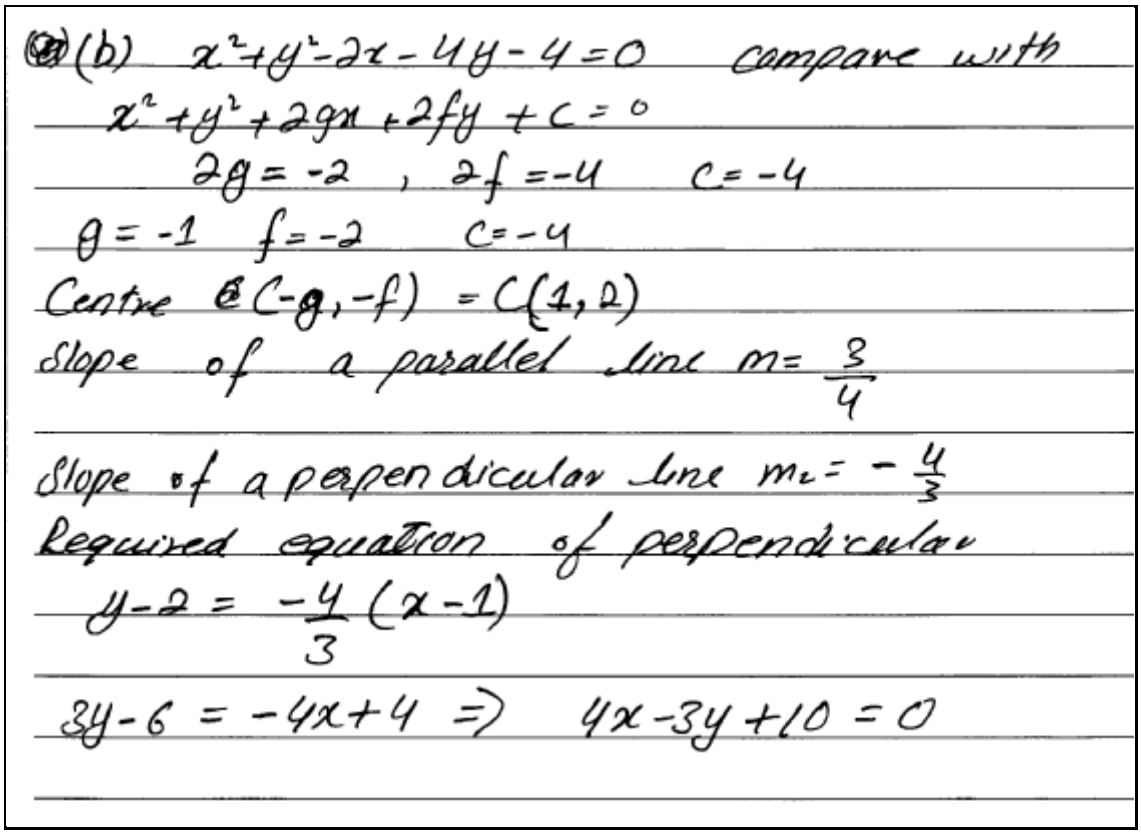


Question:6b:

Find equation of the tangent to the circle $x^2 + y^2 - 2x - 4y - 4 = 0$ which is parallel to the line $3x - 4y - 1 = 0$.

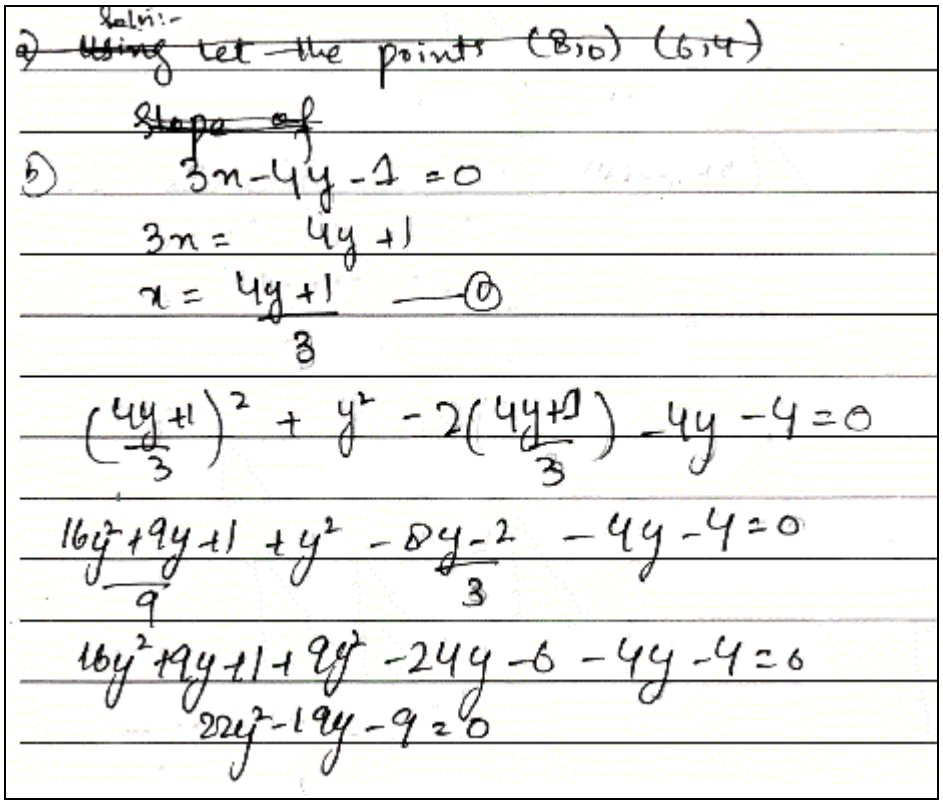
Better responses reflected correct use of general equation of circle to find the radius and centre. These values were substituted in the formula of distance of a point from a line to find the value of *c*. This value, together with gradient, was used to find the final equation.

Example:



Weaker responses reflected that most of the candidates were able to find the centre and even the radius but they did not know how to proceed further. A lot of guess work was done using the coefficients of the given equation of line to find the tangent.

Example:



Question 7:

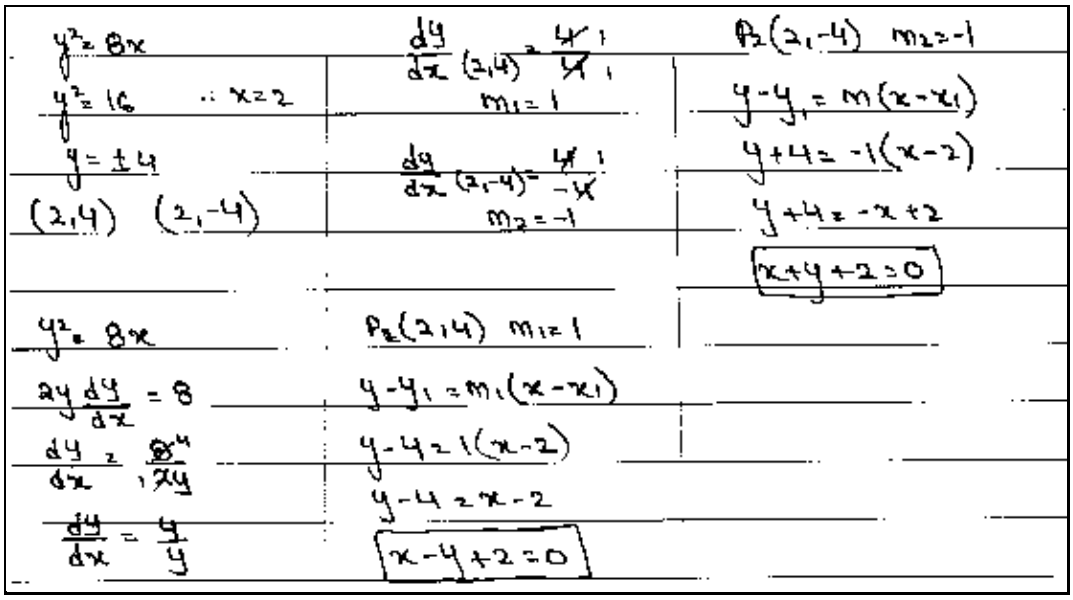
This question offered a choice to attempt any two questions from part a, b and c. Majority of the candidates attempted part b and c. All parts were well attempted.

Question 7a:

Find the equations of tangents to the parabola $y^2 = 8x$ at the points where $x = 2$.

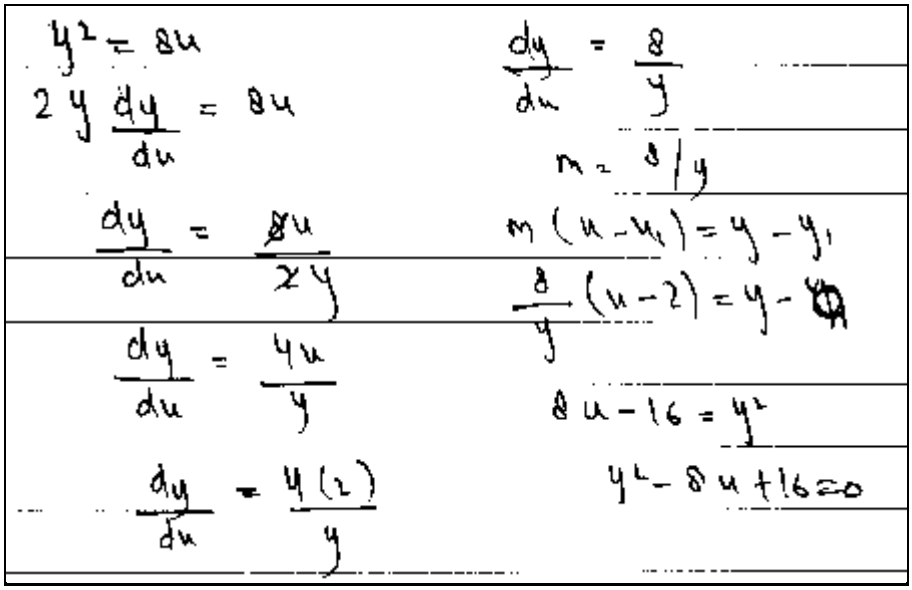
Better responses exhibited understanding of tangents to the parabola. Each point at which the two tangents touch the parabola was found. The gradient of parabola at each of the two points was found by using derivation. Correct substitutions were made to find the required two equations.

Example:



Weaker responses showed incorrect derivation of equation of parabola. Many candidates did not proceed after finding the derivative. Many candidates did not find *y*-coordinates at $x = 2$, among those who did, many ignored the fact that two *y*–coordinates are required (since it is an upward parabola, i.e. there will be two equations of tangents). Errors in substitutions were frequent as well.

Example:

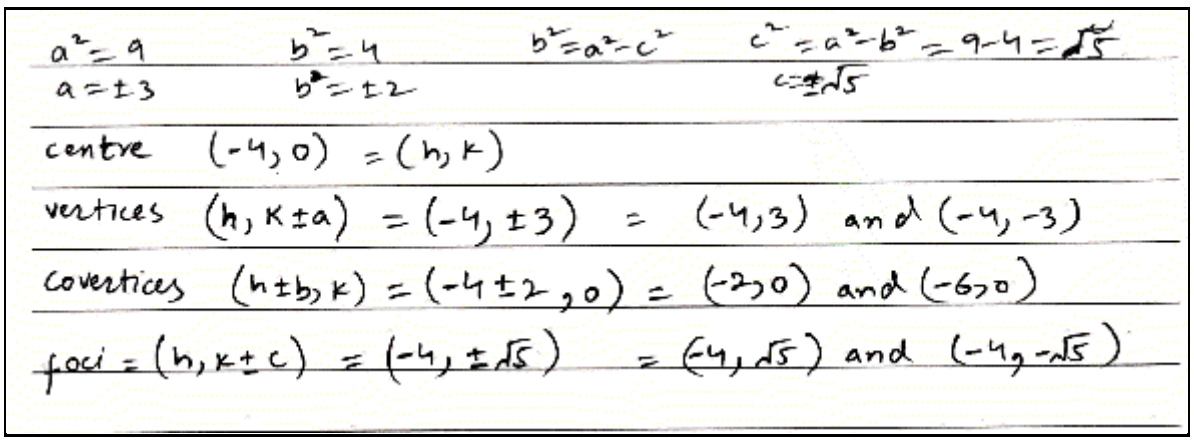


Question 7b:

Find the coordinates of centre, vertices, covertices and foci of the ellipse $\frac{(x+4)^2}{4} + \frac{y^2}{9} = 1.$

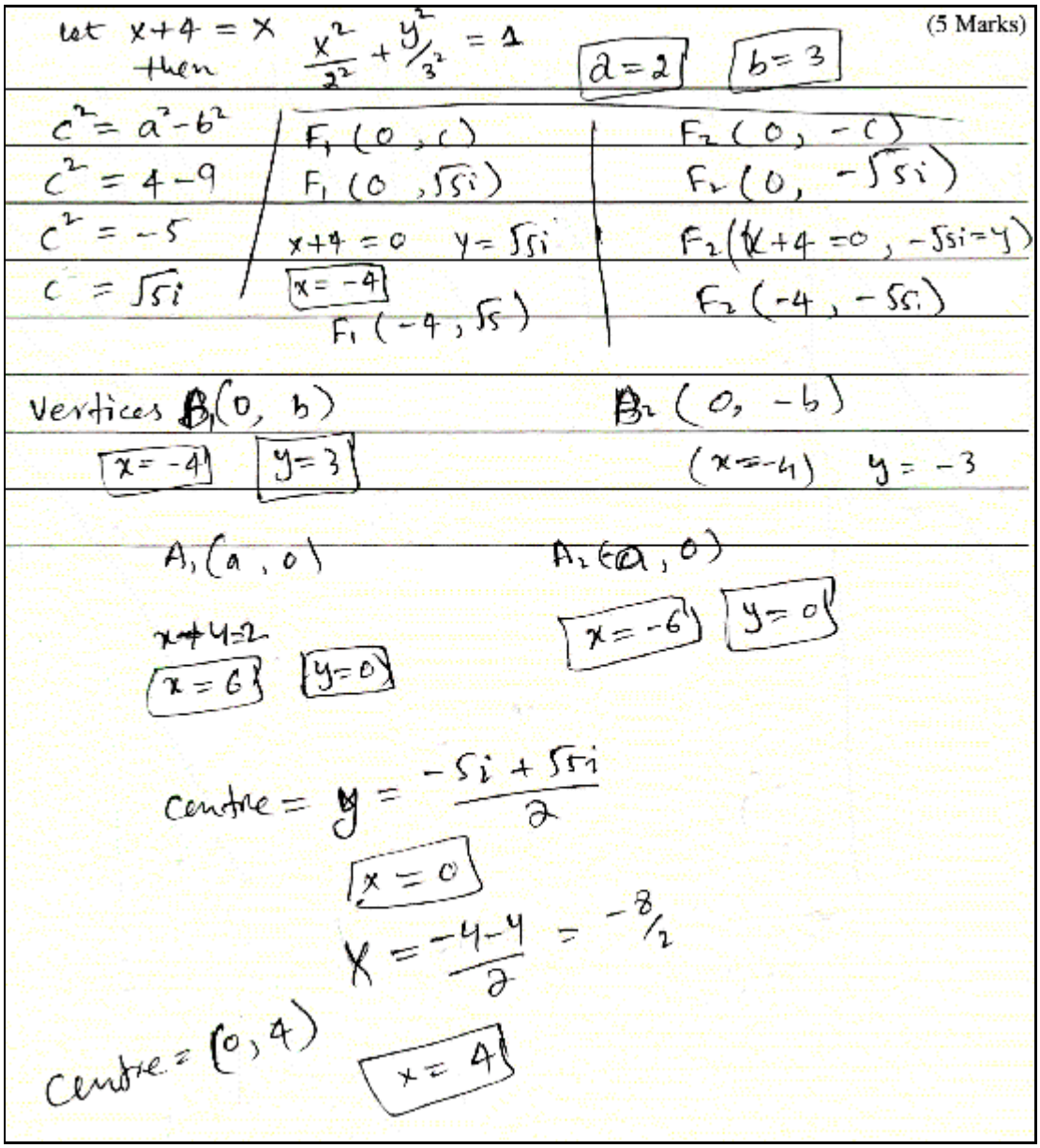
Better responses showed that candidates were able to use the information of ellipse given to find *a*, *b* and *c* using $\frac{[x-(-4)]^2}{4} + \frac{(y-0)}{9} = 1$. The vertices $(h, k \pm a)$, foci $(h, k \pm c)$ and covertices $(h \pm b, k)$ were found using correct general forms.

Example:



Weaker responses reflected weak understanding of the elements of ellipse. The candidates were not able to use the given information for their benefit. The main reason was their confusion about what *a*, *b* and *c* stand for. Since the given ellipse was translated in *x*-axis only, and not in *y*-axis, this proved to be a challenge for candidates. Incorrect formulae were frequently used.

Example:

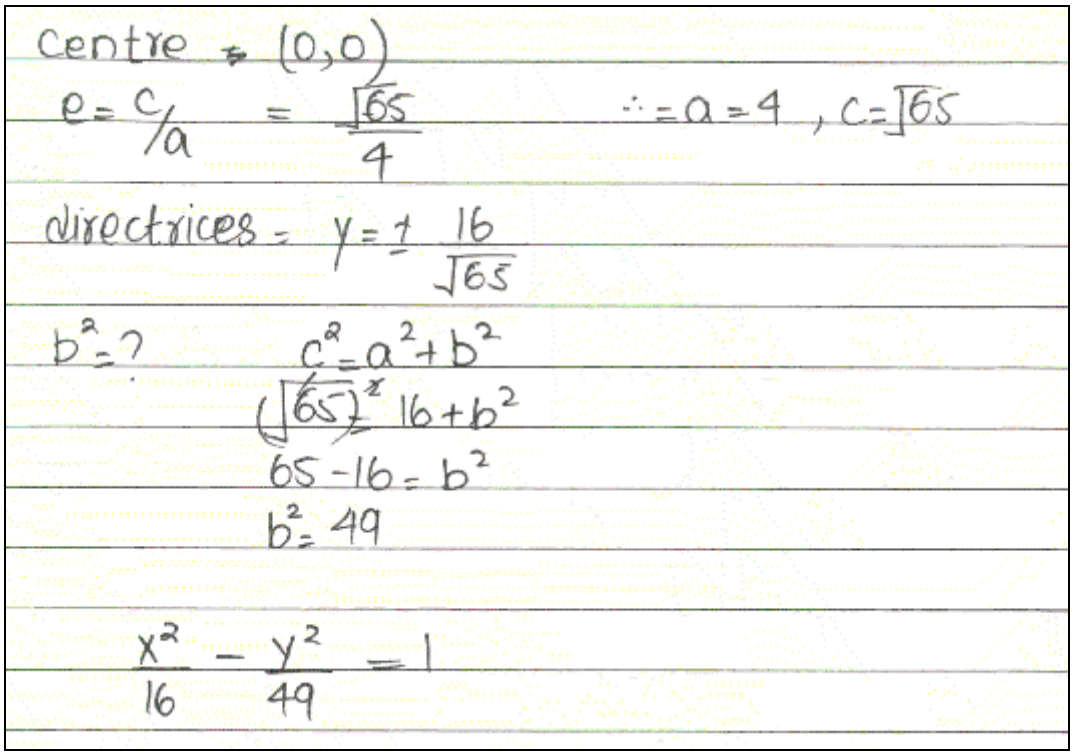


Question 7c:

Find the equation of hyperbola that has its centre at origin. Its eccentricity is $\frac{\sqrt{65}}{4}$ and its directrices are $y = \pm \frac{16}{\sqrt{65}}$.

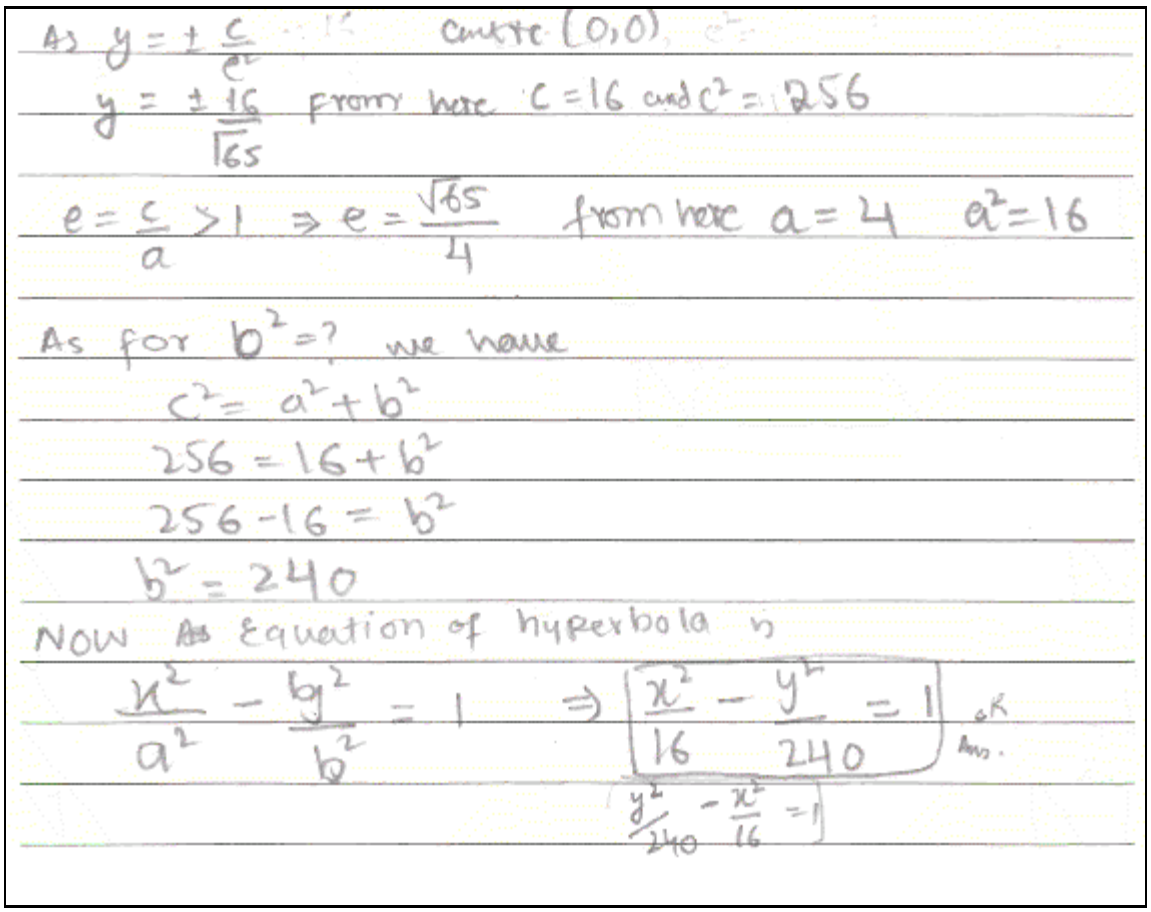
Better responses exhibit correct understanding of the elements of hyperbola. The given data was used to find the values of *a*, *b* and *c*. Candidates were able to understand that the transverse axis is along *y*-axis, hence, the general form of this hyperbola must be $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$. Correct substitution was made to find the final equation.

Example:



Weaker responses reflected weak concepts of the elements of hyperbola. They could not translate the given information in the question into an equation. Most of the candidates were able to find the values of *a*, *b* and *c* but could not identify the correct transverse axis for the hyperbola.

Example:



Question 8:

This question offered a choice between part a and b. Majority of the students chose part b as compared to part a which was also attempted much better than part a. Part a showed poor performance.

Question 8a:

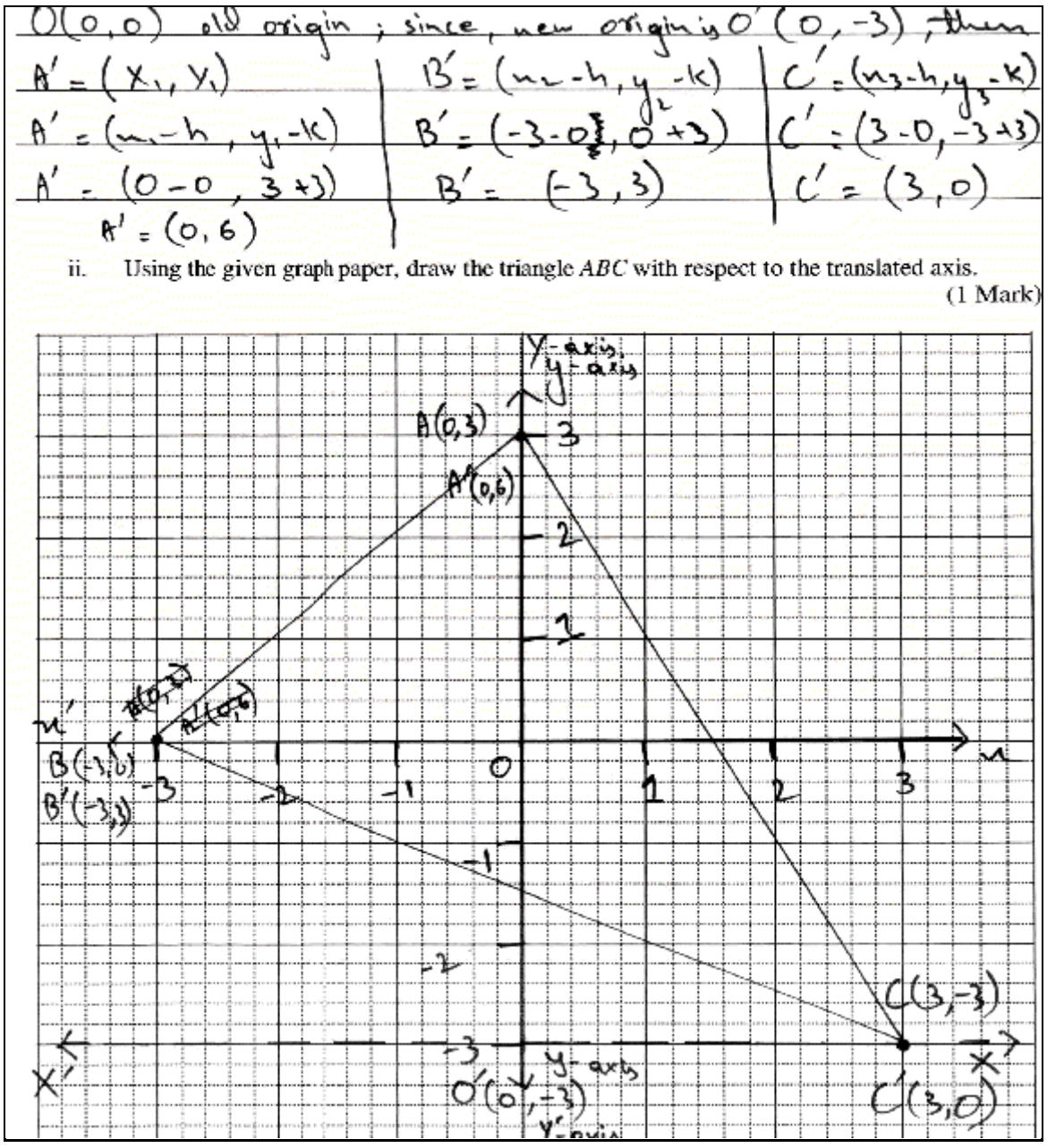
A triangle *ABC* located on *xy*-coordinate axes has vertices *A* (0, 3), *B* (– 3, 0) and *C* (3, – 3). The *x*-axis is translated through a distance of – 3 units with respect to origin while there is no change in the *y*-axis.

i. Find the new vertices of the triangle *ABC* with respect to the translated axis.

ii. Using the given graph paper, draw the triangle *ABC* with respect to the translated axis.

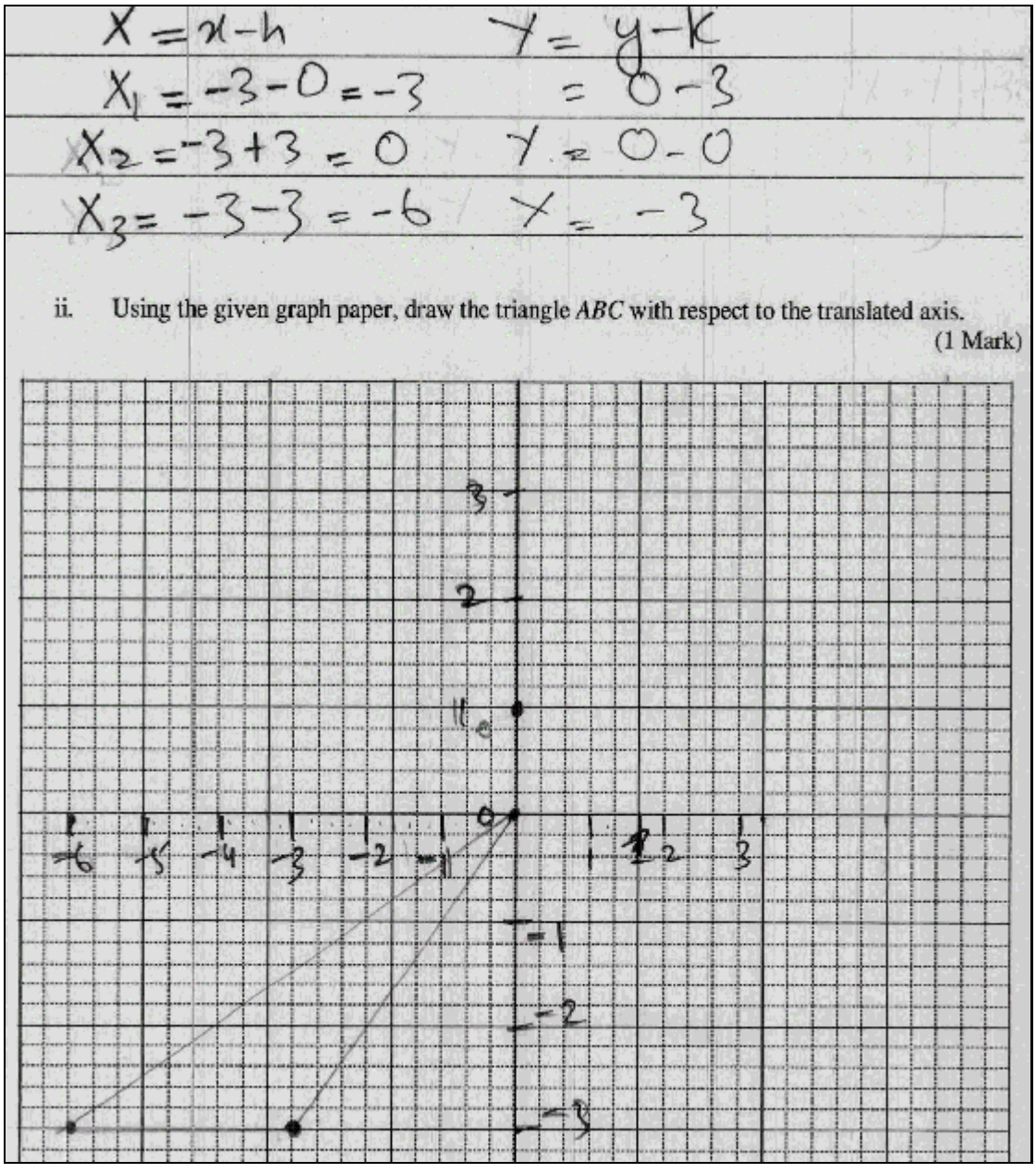
Even though it was a repeated question from past papers, it was not well attempted. There were very few *good responses* where the candidates understood that if '*x*-axis is translated through a distance of -3 units while there is no change in the *y*-axis' this means that the movement of the triangle will be + 3 units along *y*-axis. Since axes are translated through (0, –3), therefore A (0, 3) becomes $A'(0, 6)$, $B(-3,0)$ becomes $B'(-3, 3)$ and $C(3, -3)$ becomes $C'(3, 0)$. The new points were plotted on the graph to show the transformation.

Example:



Weaker responses showed confusion of candidates in using equations of transformation. The most frequent error was using *x* instead of *X* and *y* instead of *Y* in the equations of transformation. Instead of using $x = X + h$ and $y = Y + k$, candidates used $x = X - h$ and $y = Y - k$.This confusion between which variable denotes old axes and which variable denotes new axes resulted in loss of marks.

Example:



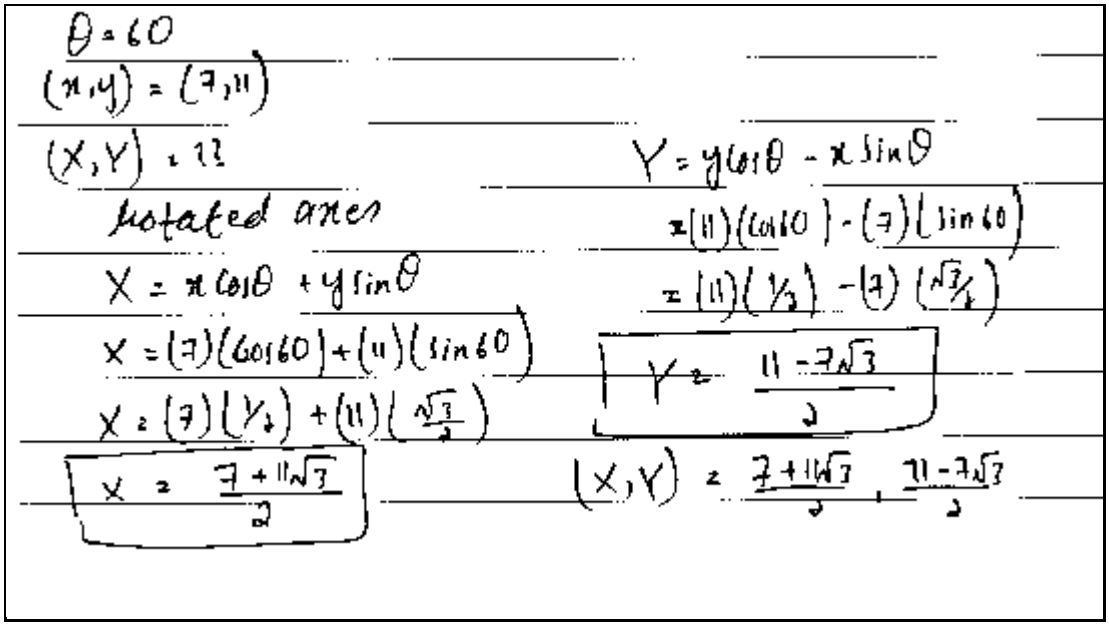
Question 8b:

The *xy*-coordinate axes are rotated about the origin through an angle of $60°$.

The co-ordinates of point *A* are (7, 11) with respect to *xy*-coordinate axes. Find coordinates of *A* with respect to the rotated axes.

Better responses exhibited the correct use of equations of transformation to get equations in terms of *X* and *Y*. Correct substitutions were made in $X = x\cos\theta + y\sin\theta$ and $Y = y\cos\theta - x\sin\theta$ to get the new coordinates of *A*.

Example:



Weaker responses reflected that candidates mostly used incorrect equations of substitution. The most frequent incorrect equations used were $X = x\cos\theta - y\sin\theta$ and $Y = y\cos\theta + x\sin\theta$ which lead the candidates to incorrect answers.

Example:

